

**Designing a Manufacturing Process for the
Production of Amoxicillin in Australia**

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On my honor as a University student, I have neither given nor received unauthorized aid on this assignment as defined by the Honor Guidelines for Thesis-Related Assignments.

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1. Executive Summary

Amoxicillin is a critical antibiotic to public health. However, this antibiotic is not produced domestically in Australia, a country which imports 90% of its medicines and is reliant on production in China and India. Local production of amoxicillin would enhance pharmaceutical self-sufficiency, improve supply security, and support national Australian objectives related to healthcare resilience. This report details a biocatalytic process to manufacture amoxicillin, with the goal of reaching supply chain independence of the antibiotic in Australia.

Our process begins with a series of four CSTRs, that react 6-APA and HPGME, our two main reactants, into amoxicillin and methanol. To reduce the amount of water in the downstream process, we then pass the product through a reverse osmosis filter. The product then passes to a series of crystallizers, and the amoxicillin is centrifuged to remove it from the liquid and dissolved initial reactants. To finish purification, the amoxicillin is washed with an ethanol solution and dried to yield a purified active pharmaceutical ingredient. Finally, the amoxicillin passes through a tableting process, where excipients are added, and it is packaged and bottled to be sold. The process converts 7.6 kg/h of 6-APA and 6.5 kg/h of HPGME to 11.95 kg/h of amoxicillin API, and 160 million tablets a year, or 8 million 20-count bottles.

Economic analysis indicates strong viability for this process. Our IRR is 26%, and with a reasonable capital cost of \$17.2 million USD, this process should be advanced for further consideration. However, additional research on components, including possible in-house production of 6-APA, recycle streams of water and ethanol, and cGMP requirements for the pharmaceutical process is needed. These changes may have significant implications for the IRR, and may change whether or not the plant should be built.

2. Introduction

This project assesses the feasibility of establishing an amoxicillin manufacturing facility in Australia using a penicillin G enzymatic process. Amoxicillin is a widely used β -lactam antibiotic in modern healthcare that is prescribed globally to treat a broad range of bacterial infections, including respiratory tract, urinary tract, skin, and ear infections. Originally developed in the early 1970s as a semisynthetic derivative of penicillin, amoxicillin is commonly produced through a process that begins with the fermentation of *Penicillium* to generate penicillin G, which is then converted into 6-Aminopenicillanic acid. The continued clinical importance of amoxicillin can be seen through its inclusion on the World Health Organization's List of Essential Medicines, highlighting its critical role in global public health (World Health Organization [WHO], 2023).

Despite its widespread use and medical significance, the amoxicillin active pharmaceutical ingredient (API) production is heavily concentrated in China and India (Australia Medicaments of Other Antibiotics, for Retail Sales Imports by Country, n.d.). This geographic concentration has created a global dependence on imported amoxicillin, exposing healthcare systems to vulnerabilities that arise from supply chain disruptions and global health emergencies. Local production of amoxicillin would enhance pharmaceutical self-sufficiency, improve supply security, and support national Australian objectives related to healthcare resilience.

2.1 Product Description

We will be making amoxicillin trihydrate ($C_{16}H_{19}N_3O_5S \bullet 3H_2O$) which is the active pharmaceutical ingredient (API) in amoxicillin. Due to the pharmaceutical applications of amoxicillin, the purity of the final product must be very high to avoid any adverse health effects

that could occur, especially when the user's immune system is already compromised from its handling of the initial infection. Additionally, Australian standards for purity of pharmaceuticals follows the United States Pharmacopeia-National Formulary (USP) standards for amoxicillin capsules which regulates the variation that can occur between what the product is labeled as and the actual contents. This specific standard has a lower limit of 90% of what it is labeled as and an upper limit of 120% of its label (USP, 2018) and with all other inactive ingredients needed for capsuling, the purity of amoxicillin needs to be high to meet these standards. We plan to produce amoxicillin trihydrate with a purity of 98% after filtration, crystallization, centrifugation, and drying (Perveen, 2020).

Our final product will be tablets of 500mg of amoxicillin, which will be distributed to pharmacies in Australia. These tablets would be used to treat a wide variety of common bacterial infections, such as ear, nose, throat, skin, and upper respiratory infections. A discussion on drug product form choice is included in section 3.6.

2.2 Previous Work

Past syntheses of amoxicillin have primarily involved the chemical acylation of 6-aminopenicillanic acid (6-APA) with α -amino-p-hydroxyphenylacetic acid via the Dane salt route. This method requires protection of the amino group on the side-chain precursor, followed by activation and subsequent coupling with 6-APA (BenchChem, 2025). While effective, this multistep process generates significant waste and requires additional separation and purification steps (Agence nationale de sécurité du médicament et des produits de santé [ANSM], 2016). More recently, enzymatic synthesis routes have gained attention as a more environmentally friendly alternative. In these processes, 6-aminopenicillanic acid reacts with

hydroxyphenylglycine methyl ester in the presence of penicillin G acylase as a biocatalyst. This approach operates under milder conditions, reduces the need for protecting groups, and minimizes byproduct formation. Due to these advantages, the enzymatic synthesis route has become increasingly attractive for industrial application and will serve as the basis for the process design in this project.

2.3 Market and Plant Sizing

Our market size for this project is 100% of amoxicillin consumed in the country of Australia. This does not include similar medications that are prescribed and tracked separately, such as amoxicillin clavulanate, which is amoxicillin and clavulanate potassium together in one drug product. The most recent estimated population of Australia is 27,536,874 people, as of March 2025 (Australian Bureau of Statistics, 2025).

The most recent reliable data for the amount of amoxicillin dispensed in Australia is for the 2013-14 through 2016-17 years (Australian Commission on Safety and Quality in Health Care, 2018). These values are reported in defined daily doses (DDD) per 1,000 people. DDD is a measurement defined by the World Health Organization to be the assumed dose per day for a drug (ATCDDD, 2025). The DDD of amoxicillin is 1.5 g.

Over the four years reported in the data, there is an average of 5.3 DDD / 1000 people dispensed per day in Australia. Using dimensional analysis, we can find a necessary value for mass produced per year:

$$5.3 \frac{DDD}{1000 \text{ ppl day}} * (27,536,874 \text{ ppl}) \left(\frac{1.5 \text{ g}}{1 \text{ DDD}} \right) \left(\frac{1 \text{ kg}}{1000 \text{ g}} \right) \left(\frac{365.25 \text{ days}}{1 \text{ yr}} \right) = 79,960 \frac{\text{kg}}{\text{year}} \approx 80,000 \frac{\text{kg}}{\text{year}} \text{ (Eq. 2.3-1)}$$

So, our target for production will be 80,000 kg per year. In terms of 500 mg tablets, this will be 16,000,000 tablets per year.

Given our target production value of 80,000 kg per year , we know that we can run at 80% capacity and still meet our target criteria. This would mean running at full capacity for 292 days each year. So, we can use 300 days per year of continuous operation as an estimate, providing room for unplanned downtime while still meeting our target production value.

2.4 Plant Siting

The location of our manufacturing plant will be at Norfolk Avenue in the Flinders Industrial Estate in New South Wales, Australia. The zone developed by Shoalhaven City Council provides access to fully serviced industrial lots that support manufacturing and large-scale operations (*Land Sales inside Flinders Industrial Estate a Boost for Jobs | Shoalhaven City Council*, 2024). The east coast of Australia, particularly New South Wales, is a hub for pharmaceutical and large-scale manufacturing, hosting companies like Pfizer and AstraZeneca. The coastal area strengthens logistic efficiency, with access to major transportation networks and ports on Australia's eastern shore, crucial for a timely distribution of products.

2.5 Major Starting Materials

In this process, amoxicillin is produced by coupling 6-aminopenicillanic acid (6-APA) with hydroxyphenylglycine methyl ester (HPGME) using penicillin G acylase (PGA) as the biocatalyst in mild conditions. The following sections describe all starting materials with their required grades, regulatory expectations, routes of supply, and roles within the process.

2.5.1 6-Aminopenicillanic Acid (6-APA)

6-APA is the β -lactam nucleus used to construct amoxicillin. It is the structural fragment that is acylated to form amoxicillin, supplying the amine site that will be acylated (Chow et al., 2005). Industrially, it is obtained via enzymatic hydrolysis of penicillin G, sourced from *Penicillium chrysogenum* fermentation. It must be purchased as a crystalline powder at a pharmaceutical reagent purity standard $\geq 98\%$ and is relatively stable at neutral conditions, able to undergo hydrolysis under acidic or basic conditions (Diender et al., 1998). It is supplied dry, not in solution, and therefore must be dissolved and rehydrated into purified water prior to use. Careful pH control is necessary to maintain ring stability and maximize nucleophilic activity.

2.5.2 D-Hydroxyphenylglycine Methyl Ester (HPGME)

HPGME is the acyl donor that transfers the hydroxyphenylglycol side chain to 6-APA required for amoxicillin synthesis. The methyl ester transfers the acyl group to the 6-APA amino group in the enzyme-catalyzed acylation (Gonçalves et al., 2003). D-HPGME is also supplied as a dry, white crystalline powder with a chemical purity $\geq 98\%$ (Bezerra et al., 2013; *Buy D-(-)-Alpha-P-Hydroxy-Phenylglycine Methyl Ester Best Price High Purity Hot*, n.d.).

2.5.3 Penicillin G Acylase (PGA)

PGA is the biocatalyst that enables selective acyl transfer from HPGME to 6-APA, and is produced by recombinant *E. coli*. It will be supplied in immobilized form on damp, pre-hydrated agarose beads, improving stability. Enzymatic synthesis of β -lactam antibiotics has become increasingly attractive as an industrial process due to its high selectivity, reduced reaction steps, and lower toxicity of waste products per kilogram of antibiotic (Gonçalves et al., 2000).

Additionally, using enzymes such as penicillin G acylase (PGA) as catalysts is increasingly attractive due to their high specificity, selectivity, and activity in mild reaction conditions (Gonçalves et al., 2000).

3. Discussion

3.1 Reactor Design

The reactor system was designed to continuously produce 80,000 kg/yr of amoxicillin from 6-aminopenicillanic acid (6-APA) and hydroxyphenyl glycine methyl ester (HPGME). Based on this production target and reaction stoichiometry, the required feed rates to the reactor were calculated to be 6.84 kg/h of 6-APA and 5.76 kg/h of HPGME. To simplify downstream separation and minimize unreacted reagents, a target conversion of 98% with respect to 6-APA was selected as the primary performance criterion. A target conversion of above 90% is standard for commercial processes, with optimal processes aiming for above 95% (Sawant et al., 2020; Xu et al., 2018). A higher target was selected due to the high cost of 6-APA and the purity requirements of pharmaceutical production. Increasing conversion reduces unreacted intermediate, minimizing downstream separation processes. Due to the limited aqueous solubility of both 6-APA and HPGME, a relatively large volumetric flow of water was required to maintain all species in solution throughout the reactor. A volumetric flow of 10,800 kg/h of water is required to maintain all species in solution. The process was designed for continuous, steady-state operation under mild aqueous conditions consistent with industrial pharmaceutical manufacturing practices.

3.1.1 Reactor Selection

A continuous stirred-tank reactor (CSTR) was selected as the reactor type for the enzymatic synthesis of amoxicillin due to its ability to maintain uniform substrate and enzyme concentrations and provide effective temperature and mixing control in continuous operation. Continuous stirred-tank reactors are widely used in fine chemical and pharmaceutical synthesis

because they produce consistent reaction environments with well-mixed reactants, simplifying process control and scale-up relative to batch or tubular alternatives (Cherkasov et al., 2023). Unlike plug-flow reactors (PFRs), which can exhibit steep concentration gradients along the reactor length and require careful monitoring to maintain enzyme activity and prevent local substrate depletion, CSTRs maintain a homogeneous reaction mixture that is easier to control, particularly for liquid-phase enzymatic reactions. CSTRs in series can approximate plug-flow behavior through staged conversion while retaining the operational advantages of stirred tanks, such as reduced complexity, improved reliability, and easier control of residence time distribution. In this operation, aqueous feed containing 6-APA, HPGME, enzyme, and solvent continuously enters the first stirred tank with rapid mixing to ensure uniform concentrations. The reaction mixture will flow sequentially through each reactor in cascade, with partial conversion occurring in each stage. Product and unreacted species will exit the final reactor at steady state, with the staged configuration allowing for high overall conversion.

3.1.2 Kinetic Considerations

The enzymatic synthesis of amoxicillin was modeled using Michaelis–Menten kinetics with competitive inhibition to describe the reaction dynamics of both the primary synthesis and side reactions, consistent with previously published models of this system (Neto et al., 2025). Two reaction steps were considered, ester consumption and amoxicillin formation, each governed by rate expressions that account for substrate and product inhibition. Enzyme concentration was expressed in international units per liter (IU/L), reflecting standard biopharmaceutical reporting practices and allowing direct use of literature activity data. Several simplifying assumptions were made to enable reactor design: enzyme deactivation was

neglected, mass transfer limitations were assumed negligible due to turbulent mixing, and temperature and pH were assumed constant throughout the reactor. The kinetic parameters used in this study, including k_{cat} , K_m , and inhibition constants, are summarized in Appendix 10.1, with the complete rate law expressions. These assumptions and models are reasonable for continuous, well-mixed CSTR operation and allow the kinetic model to be directly coupled with material balances for reactor sizing and design.

3.1.3 Methodology Using MATLAB

Because the system operated under kinetic control, reaction rates were determined using the Michaelis-Menten expressions described above. These rate expressions were coupled with steady-state material balances for a continuous stirred tank reactor (CSTR) to determine outlet concentrations for a given reactor volume. To achieve the target 98% conversion of 6-APA, a cascade of CSTRs was modeled in MATLAB. The total reactor volume and number of stages were systematically varied, and the resulting conversions were evaluated to identify the minimum reactor volume and number of tanks required to meet the conversion constraint.

3.1.4 Reactor Configuration

The final reactor design consists of 4 cascading CSTRs operated in a series, with a total reactor volume of 248.7 L and an individual reactor volume of 62.2 L. This staged CSTR configuration was selected to achieve the required 98% conversion of 6-APA while maintaining the operational advantages of well-mixed reactors. By distributing the reaction across multiple smaller tanks, higher overall conversion was obtained compared to a single CSTR of equivalent total volume, while avoiding the complexity associated with alternative reactor types. Each

reactor operates at steady state with a total residence time of 1.38 minutes, providing sufficient contact time for the enzymatic reaction while minimizing enzyme exposure to prolonged shear and thermal stress. Using these kinetic models and material balances, the reactor cascade was sized to meet the target conversion while maintaining practical operational conditions.

3.1.5 Impeller and Baffle Design

Each of the four CSTRs in the cascade is designed with a tank diameter of 0.38 m and a height of 0.56 m to accommodate the required reactor volume while maintaining appropriate aspect ratios for effective mixing. A 45° pitched-blade turbine will be used as the impeller to ensure uniform mixing throughout the tank. The impeller has a diameter of 0.11 m and is operated at 380 rpm, providing a power input of 11.7 kW, sufficient to achieve turbulent flow and minimize mass transfer limitations. To further enhance mixing and prevent vortex formation, four vertical baffles are included in each reactor, each with a width of 3.8 cm and a thickness of 0.4 cm. This combination of impeller geometry, speed, and baffles is expected to maintain a homogeneous reaction environment, supporting consistent enzyme activity and maximizing conversion in each stage of the CSTR cascade.

3.1.6 Enzyme Use

The enzyme used in the reactor system is penicillin G acylase (PGA), immobilized on agarose beads as described previously. Each reactor contains 50,000 IU/cm³ of enzyme activity, and the immobilized enzyme is replaced every two weeks to align with the scheduled downtime of the other unit operations (Bernegger-Egli & Witholt, 1999). The reactors are designed with sufficient agitation to maintain the agarose beads in suspension and ensure adequate mass

transfer throughout the reaction medium. To prevent bead loss between reactors and downstream equipment, internal retention screens or membrane filtration systems are used to retain the immobilized enzyme beads within each reactor while allowing the liquid process stream to flow continuously through the system.

3.2 Reverse Osmosis Design

Because of the low solubility of our reactants in water, and the fact that reaction data are only published in water, our process requires a high quantity of water to enter and exit the reactor (10,600 kg/h). We need an intermediate process to remove a large quantity of water, while conserving almost all of our amoxicillin product. So, a reverse osmosis unit operation was selected.

Unlike a traditional reverse osmosis process, used in water purification, we will be capturing the retentate, or the liquid that does not pass through the membrane, which will be rich in amoxicillin and other impurities. The permeate, or the liquid that does pass through the membrane, will be low in non-water compounds. This use of filtration is common in concentrating biomolecules such as proteins, but due to the relatively low molecular weight of amoxicillin, it will pass through most common ultrafiltration systems, necessitating nanofiltration instead.

To design this process, we selected a target exiting concentration of 10 g/L of amoxicillin in water to enter the downstream process. This is a concentration increase of roughly 10x, meaning that about 90% of the water will be removed, reducing the strain on downstream processes. Additionally, because of the low concentration of amoxicillin (1.35 g/L), we concluded that concentration polarization can be neglected, simplifying our design equations.

For our reverse osmosis membrane, we selected the Dupont Filmtec BW30FR-400/34 RO Membrane (*Dupont Filmtec Compatible Membranes*, n.d.). This membrane was primarily selected due to its large salt rejection rate and average permeate, relative to its area. Its specifications are listed below in Table 3.2-1.

Table 3.2-1. Specifications of the Dupont Filmtec BW30FR-400/34 RO Membrane.

Parameter	Value
Salt Rejection Rate	99.5%
Average Permeate	10,500 GPD @ 225 PSI (15.5 bar)
Maximum Operating Pressure	600 PSI (41.4 bar)

Because amoxicillin has a higher molecular weight than salt, amoxicillin will be rejected at a higher rate than salt in the membrane. Since we cannot calculate or estimate the true rejection rate, we will assume that the rejection rate of amoxicillin, 6-APA, and HPGME are all equal. Further, we will assume that their rejection rate is 99.5%, as this would result in the most loss of product and is a “worst-case scenario”. Additionally, we select an operating pressure of 10 bar gauge.

With a rejection coefficient and a target retentate concentration, we can then calculate key parameters for the reverse osmosis process. Since we are neglecting concentration polarization, our primary design equation is as follows:

$$u_p = \frac{Q_p}{A} = K(\Delta P - RTC_f) \text{ (Eq. 3.2-1)}$$

To solve for K, which is in units of $\text{m s}^{-1} \text{Pa}^{-1}$, we use the specifications from the membrane with the following dimensional analysis:

$$K = \left(\frac{10,500 \text{ GPD}}{225 \text{ PSI} \times 0.21 \text{ m}^2}\right) \left(\frac{1 \text{ day}}{86400 \text{ s}}\right) \left(\frac{1 \text{ m}^3}{264.2 \text{ gal}}\right) \left(\frac{1 \text{ PSI}}{6895 \text{ Pa}}\right) = 1.41 \times 10^{-9} \text{ m s}^{-1} \text{Pa}^{-1} \text{ (Eq. 3.2-2)}$$

Then, we can solve for the flowrates of each component. For amoxicillin, 6-APA and HPGME, this is simple; since we are assuming a rejection of 99.5%, 99.5% of each component will be retained, and the remaining 0.5% will permeate. For the two liquids, this is more complicated.

To solve for Q_p , we can use the fact that we fixed our permeate concentration at 10 grams of amoxicillin per liter (or kg) of water. Because 14.6 kg/h of amoxicillin enters the process, 99.5%, or 14.5 kg/h of amoxicillin, exits. Therefore, our permeate flow of water is 1450 kg/h of water. To complete the material balance, the remainder, or 9150 kg/h of water, exits as permeate. For methanol, we can assume that the same proportion as water exits in each stream, meaning 0.15 kg/h of methanol is retained, and 0.95 kg/h of methanol permeates.

Then, we can solve for permeate flux (u_p), using the known feed concentration of 1.37 g/L, to get $u_p = 1.41 \times 10^{-3} \text{ m s}^{-1}$, or $1.41 \text{ L m}^{-2} \text{ s}^{-1}$. This value, with Q_p of 9150 kg/h, allows us to solve for the required area of 6.5 m^2 .

Below is a table of the amount of material that enters the RO membrane, permeates through, and is retained.

Table 3.2-2. Material balances in the reverse osmosis process.

Component	Flow in (kg/h)	Permeate flow (kg/h)	Retentate flow (kg/h)
Water	10600	9150	1450
Methanol	1.1	0.95	0.15
Amox.	14.6	0.07	14.5
6-APA	0.19	0.001	0.19
HPGME	0.12	0.001	0.12

3.3 Crystallization Design

The crystallization system was designed to recover 80,000 kg/yr of amoxicillin from the reactor effluent through controlled pH adjustment and temperature reduction to decrease its aqueous solubility. Because amoxicillin is an amphoteric compound containing both acidic and basic functional groups, its solubility is strongly dependent on solution pH, with minimum solubility occurring near its isoelectric region (Sinko, 2011). Selective crystallization was achieved by adjusting the solution to conditions that minimize amoxicillin solubility while maintaining 6-aminopenicillanic acid (6-APA) and hydroxyphenylglycine methyl ester (HPGME) in solution. Hydrochloric acid (30 wt%) and aqueous ammonia (25 wt%) were selected as pH-adjusting agents due to their widespread industrial use, compatibility with pharmaceutical processing, and ease of downstream removal.

Temperature reduction was accomplished using jacketed crystallizers cooled with ethylene glycol to provide controlled supersaturation while avoiding excessive nucleation. Mixed-suspension mixed-product removal (MSMPR) crystallizers were selected for this process because they are widely used in continuous crystallization and provide steady-state operation with predictable crystal size distributions (McCabe et al., 2005). Operation of MSMPR units in series further enhances recovery by allowing staged supersaturation control and increased overall crystallization yield while maintaining impurity solubility.

3.3.1 Crystallizer Selection

A mixed-suspension mixed-product removal (MSMPR) crystallizer was selected for this process due to its suitability for continuous operation and its ability to produce a controlled and predictable crystal size distribution under steady-state conditions. MSMPR crystallizers are

widely used in pharmaceutical manufacturing because they provide uniform supersaturation throughout the vessel, allowing consistent nucleation and growth behavior (McCabe et al., 2005). Continuous operation aligns with the upstream reactor design and supports steady production of 80,000 kg/yr of amoxicillin without the scheduling constraints associated with batch processing.

Alternative crystallizer configurations were evaluated, including batch cooling crystallizers and draft-tube-baffle (DTB) crystallizers. Batch crystallization was considered due to its widespread use in pharmaceutical manufacturing and its flexibility in handling variable operating conditions. However, batch systems introduce production downtime, increased labor requirements, and greater variability in crystal size distribution, making them less desirable for an integrated continuous process. Draft-tube-baffle crystallizers were also considered because of their ability to promote classified growth and reduce fine generation. However, DTB units are more complex, require higher capital investment, and are typically favored for larger crystal production or high-throughput commodity applications. Given the moderate production scale and the need for steady, controllable operation, an MSMPR crystallizer provided the most practical balance between operational simplicity, capital cost, and product quality. To increase overall recovery while maintaining impurity solubility, multiple MSMPR units were arranged in series. Staging allows supersaturation to be distributed across multiple vessels, improving yield while limiting excessive nucleation and promoting controlled crystal growth.

3.3.2 Thermodynamic and Solubility Considerations

Amoxicillin trihydrate is an amphoteric compound containing both acidic and basic functional groups, resulting in strong pH-dependent solubility behavior. Amphoteric pharmaceuticals exhibit minimum solubility near their isoelectric point, where the net molecular

charge approaches zero and electrostatic interactions with water are minimized (Sinko, 2011). The isoelectric point of amoxicillin is approximately pH 4.7, and operation near this pH was therefore selected to minimize solubility and promote crystallization. By adjusting the reactor effluent to this region, supersaturation can be generated selectively for amoxicillin while maintaining favorable solution conditions for impurity retention. In addition to pH effects, amoxicillin solubility decreases with decreasing temperature, consistent with typical solid–liquid equilibrium behavior for organic compounds (McCabe et al., 2005).

Controlled cooling was therefore implemented to further reduce solubility and increase supersaturation in a regulated manner. The combined use of pH adjustment and temperature reduction allows independent control over thermodynamic driving force and nucleation rate, improving recovery while limiting excessive fine formation. The primary impurities present in the slurry, 6-aminopenicillanic acid (6-APA) and hydroxyphenylglycine methyl ester (HPGME), also exhibit pH- and temperature-dependent solubility. However, due to their relatively low concentrations in the reactor effluent and the large volumetric flow of water required for dissolution upstream, the system does not reach supersaturation with respect to these species under the selected operating conditions. As a result, selective crystallization of amoxicillin is achieved while maintaining impurities in solution.

3.3.3 Supersaturation and Nucleation Control

Supersaturation is the thermodynamic driving force for crystallization and is defined as the difference between the bulk solute concentration and the equilibrium solubility at a given temperature and pH. In this process, supersaturation was generated through a combination of controlled pH adjustment to the isoelectric region and gradual cooling within the crystallizer.

Maintaining moderate supersaturation levels was critical to promote crystal growth while avoiding excessive primary nucleation. In continuous MSMPR crystallizers, nucleation and crystal growth occur simultaneously under steady-state conditions. High supersaturation favors rapid nucleation and the formation of fine particles, whereas lower, controlled supersaturation promotes growth of existing crystals. Excessive fines can significantly reduce filtration efficiency and increase downstream processing time. Therefore, staged crystallization was implemented to distribute the overall driving force across multiple vessels, reducing local supersaturation and promoting more controlled crystal growth. By regulating both pH and cooling rate, the crystallization system was designed to balance recovery and slurry handling properties while maintaining operational stability. This approach supports efficient downstream solid–liquid separation without requiring complex crystal classification equipment.

3.3.4 Mass Balances and Design Methodology

The crystallization system was modeled in MATLAB as a series of three steady-state MSMPR crystallizers operating at progressively lower temperatures and controlled pH conditions. A stagewise equilibrium approach was used in which amoxicillin crystallization in each vessel was determined by comparing the bulk concentration to the equilibrium solubility at the specified temperature and pH. If the bulk concentration exceeded the solubility limit, the excess mass was assumed to crystallize until the dissolved concentration reached equilibrium. For each stage, the dissolved amoxicillin concentration was calculated based on the updated liquid volume, accounting for water added through acid or base addition. Solubility values were specified for each operating condition, and the mass of crystals formed was determined from the difference between inlet mass flow and the equilibrium dissolved mass. This approach assumes

ideal mixing and that sufficient residence time is provided for the liquid phase to approach saturation conditions. pH adjustment requirements were calculated using molar hydrogen ion balances. The required moles of acid or base were determined from the change in hydrogen ion concentration corresponding to the target pH of each stage. Mass flow rates of 30 wt% hydrochloric acid and 25 wt% aqueous ammonia were then computed based on molecular weights and solution composition. The additional water introduced through these solutions was incorporated into the stage mass balance. An energy balance was performed for each crystallizer to determine cooling requirements. The total heat duty included both sensible cooling of the liquid stream and the heat of crystallization of amoxicillin. The sensible heat removal was calculated using mixture heat capacity and temperature change, while the crystallization contribution was determined from the specified enthalpy of crystallization. The resulting heat duty was used to size the chilled water and ethylene glycol flow rates required for jacket cooling. Crystallizer volume was determined using the residence time relationship:

$$\tau = \frac{V}{Q} \text{ (Eq. 3.3.4-1)}$$

where the residence time was selected as 2 hours to allow sufficient time for equilibration and crystal growth. The working volume of each MSMPR was calculated based on slurry mass flow rate and density. Vessel dimensions were estimated assuming a height-to-diameter ratio of 1.5, consistent with standard stirred-tank design practice (McCabe et al., 2005). This staged equilibrium-based methodology provided estimates of crystallization yield, cooling duty, and equipment size while maintaining consistency with steady-state MSMPR design principles.

3.3.5 Configuration and Cooling Design

The crystallization system consists of three MSMPR crystallizers arranged in series, each with a working volume of 2.93 m³. Based on the selected 2-hour residence time and slurry volumetric flow rate, this volume provides sufficient time for equilibration and crystal growth under steady-state operation. Each vessel was designed with a diameter of 1.36 m and a height of 2.03 m, corresponding to a height-to-diameter ratio of approximately 1.5. This aspect ratio is consistent with standard stirred-tank design practice and promotes effective suspension of crystals while maintaining manageable mechanical dimensions.

Each crystallizer is equipped with a 45° pitched-blade turbine impeller to ensure adequate suspension of solids and uniform supersaturation throughout the vessel. The impeller diameter is 0.45 m and operates at 90 rpm, delivering 18.5 kW of power. This configuration maintains turbulent flow conditions within the tank, minimizing concentration and temperature gradients while preventing crystal settling.

The three crystallizers operate at progressively decreasing temperatures and controlled pH conditions to generate staged supersaturation. The first crystallizer operates at 10°C and pH 1.5 with a calculated heat removal requirement of 26 kW. The second stage operates at 8°C and pH 3.5 with a heat duty of 3.4 kW, and the final stage operates at 5°C and pH 4.7 with a heat duty of 5.1 kW.

The larger duty in the first stage reflects the greater sensible cooling load and initial crystallization heat release. Cooling is provided through external jackets using ethylene glycol as the heat transfer fluid. Ethylene glycol enters the jacket system at -3°C at a total flow rate of 7,570 kg/h and exits at 7.8°C. This temperature rise provides sufficient driving force for heat

removal while avoiding freezing risk and maintaining stable operation. The selected coolant flow rate ensures that the required heat duties are met while maintaining practical jacket heat transfer coefficients and safe operating temperatures.

3.4 Centrifugation

The centrifuge is intended to remove most of the liquid, including dissolved components, from the system to allow for a wetcake to advance in the process towards washing and drying. In order for the process to remain continuous, we selected a disc stack centrifuge, as described below.

First, we calculate a terminal velocity of amoxicillin crystals:

$$V_g = \frac{d_p^2 (\rho_p - \rho_f) g}{18\mu} \quad (3.4-1)$$

where V_g is terminal velocity, d_p is the diameter of the particle (here, 150 μm), ρ is the density of the particle (1.4 g/mL) and the fluid (1.0 g/mL), and μ is the viscosity of the liquid (0.001 Pa s). The fluid was assumed to be pure water, due to its very low methanol content. This calculation gave a terminal velocity of $1.06 \times 10^{-7} \text{ m s}^{-1}$.

Given our water flow rate at this stage is 1,149 kg/s of water, or $3.2 \times 10^{-4} \text{ m s}^{-1}$, we can divide by the terminal velocity to get an equivalent settling pond area Σ of 3000 m^2 . This will be used in the design equation for a disc stack centrifuge:

$$\Sigma = \frac{2\pi(n-1)\omega^2}{3g} \cot(\Theta) (r_o^3 - r_i^3) \quad (3.4-2)$$

Parameters, variables in this equation, and values are included in Table 3.4-1.

Table 3.4-1. Parameters of the disc stack centrifuge.

Parameter	Variable in Eq. 3.4-2	Value
Inner radius	r_i	25 cm
Outer radius	r_o	45 cm
Discs	n	50
Angle	θ	45°
Rotational speed	ω	600 rpm
Power consumption	N/A	130 W

From these parameters, we can also calculate the power consumption of the centrifuge while in operation to be 130 W.

3.5 Washing and Drying

3.5.1 Washing

The washing system was designed to continuously wash the Amoxicillin wet cake to remove the HPGME, ammonia, and HCl still present after crystallization and centrifugation with a final output of 11.95 kg/h of amoxicillin or approximately 80,000 kg of annually. The washing liquid is a 30% ethanol solution by weight with a total flow rate of 7.16 kg/h.

The washing step consists of one rotary drum filter. A rotary drum filter was selected for this process as they are commonly used for continuous displacement washing in pharmaceutical industries. The rotary drum filter will utilize an approximate volume/volume wash ratio of 2 L washing liquid/1 L wet cake liquid volume. This ratio was selected as wash ratios between 1.5-2 tend to remove the highest concentration of impurities per volume wash (Ruslim et al., 2007). They will operate at 25°C, which was selected due to solubility constraints of HPGME and amoxicillin.

The flow rate and ethanol composition of the washing liquid was dictated by the solubility of amoxicillin and HPGME in water and ethanol solutions. While HCl and ammonia also need to be removed, both are very soluble in a 30% ethanol solution, 0.585 g/g solvent and 0.367 g/g solvent respectively, and will therefore not determine the flow rate. However both amoxicillin and HPGME are very slightly soluble in both ethanol and water. Removing all HPGME from the wet cake will compete with the losses associated with the solubility of amoxicillin in the same solution. At lower ethanol compositions and temperatures, amoxicillin is more soluble in solution than HPGME, making the removal of all HPGME while still maintaining an output flow rate of 11.95 kg/h amoxicillin impossible. However at 25°C, a 30 wt% ethanol solution will dissolve more HPGME per gram of solvent than amoxicillin, with

0.00319 g amoxicillin/g solvent and 0.004388 g HPGME/g solvent (Felix et al., 2016; Hu et al., 2016). Solubility calculations can be found in Appendix 10.2.

Using a 30 wt% ethanol solution at 25°C, the expected loss of amoxicillin to solubility is 0.17% of the final product. To account for losses of amoxicillin during washing, the flow rate of amoxicillin into the washing stage was increased to 11.97 kg/h. The washing flow rate of 7.2 kg/h was determined by solving a system of linear equations representing the material balance around the washing stage. The system of equations can be found in Appendix 10.3.

3.5.2 Drying

The drying stage was designed to remove all remaining liquid from the wet cake after washing to produce pure amoxicillin that can be used for tableting. This stage consists of a single vacuum tray dryer and is operated as a batch process. Each batch will produce 75 kg of amoxicillin to match the batch size of the tableting process. It will take 6.3 hours to accumulate 75 kg of amoxicillin from the washing stage. Along with the amoxicillin, 4.5 kg of ethanol and 18.7 kg of water will also enter the drying stage.

The vacuum tray dryer was selected as it allows for the removal of the entrained liquid at lower temperatures than their atmospheric boiling points. At higher temperatures, amoxicillin will degrade so maintaining moderate temperatures is imperative for the drying of amoxicillin. The model of dryer for this process is the FZG-20 which can dry batches with a maximum drying capacity of 120 kg. It uses 48 trays for drying with a total drying surface area of 14.13 m². It will be operated at 35°C and 0.033 atm. The heating power of it is 21 kW which is provided by low-pressure steam.

At 0.03 atm, the heat of vaporization of a 30 wt% ethanol solution can be approximated to be 1990.2 kJ/kg. Removal of all ethanol and water from the wet cake will require an input of 46,200 kJ. Using the heating power of 21 kW, the time for each batch of drying will take 37 minutes. The evaporated water and ethanol will be condensed using 10°C cooling water in a condenser before being sent to waste water treatment.

3.6 Product Formulation

Patients cannot just be sent amoxicillin in powder form; the active pharmaceutical ingredient (API) must be made into a product before it can be sold. In this section, we describe how we decided on the form of drug product, how we created our batch size, an overview of the tableting process, and a note on re-use of ethanol in tablet production.

3.6.1 Drug Product Selection

Our first decision on product formulation was the type of drug product we would produce. Standard drug products include tablets and capsules, but amoxicillin is also produced as a chewable tablet and a liquid suspension, for pediatric uses (NIH MedlinePlus, n.d.). We decided that we would only be producing one of these forms to reduce cost of the drug product formulation process, and because less-used forms such as liquid suspensions, because they are rarer, can be imported without significant supply chain risk, or produced locally in small amounts at contracted manufacturing organizations, if necessary.

We chose to move forward with a tableting process. Some information on industrial encapsulation processes is available, but this often fails to include important information, including excipient concentrations in capsules.

3.6.2 Batch Sizing

As seen below in the process overview, our process takes a defined period of time regardless of batch size, due to its recipe-based nature. We estimate our batch size by the estimated time of the batch, assuming that it is best if batches are not being processed simultaneously to reduce employee cost and assure good quality control.

We estimate that between the drying step discussed above, and the tableting process below, the process will take approximately 7 hours. So, we can complete three batches per day while the plant is in operation. However, this does not mean that nothing else is happening during a batch. While other processes are occurring, equipment is also being cleaned and prepared for the following batch to reduce lost time.

Assuming a total production of 80,000 kg per year, this equates to 75 kg of API processed per batch. Because each tablet needs 500 mg of API, this results in 150,000 tablets per batch.

3.6.3 Tableting Process Overview

We will be using the tableting process for amoxicillin as described in *Handbook of Pharmaceutical Manufacturing Formulations*, Volume 1 (2004). This source includes quantities of excipients and the process of tableting, as included below. Note that the handbook gives excipient amounts for a 250 mg tablet, and we scaled this to work for a 500 mg tablet. Below in Table 3.6.3-1 are the amounts of components used in tableting, both on a per-tablet and per-batch basis.

Table 3.6.3-1. Ingredients used in tableting.

Material	Amount per Tablet (mg)	Amount per Batch (kg)
Amoxicillin API	500 ¹	75.0
Cellulose microcrystalline	57	8.55
Povidone	40	6.0
Magnesium stearate	7.0	1.05
Ethanol, 90%	N/A ²	8.3 ²

¹ This assumes an activity of 1000 mcg/mg.

² See Section 3.6.4 on ethanol use.

The process is completely in batch and proceeds as follows:

1. Pass amoxicillin API through a 595- μ m aperture screen.
2. Pass amoxicillin API through a Fitz mill.
3. Mix the amoxicillin with cellulose in a V mixer for 30 minutes.
4. Add 40% of the ethanol (3.3 kg) to the V mixer, mix for 15 minutes.
5. In a separate vessel, dissolve povidone in the remaining ethanol. Once dissolved, add this to the V mixer and mix for 15 minutes.
6. Pass the mixture through a 4.76-mm aperture screen.
7. Transfer the mixture to a dryer, and dry for 3 hours.
8. Pass the mixture through a 1.2-mm aperture screen.
9. Add half of the mixture to a V mixer. Pass the magnesium stearate through a 500- μ m aperture screen, and add to the V mixer. Mix for 20 minutes, then add the second half of the mixture and mix for another 10 minutes.
10. Send the tablets through a tablet press.

This recipe gives all the major pieces of equipment, but they still must be designed.

Below, we will give the design of each piece of equipment used in the tableting process.

3.6.3.1 Aperture Screens

We use four aperture screens in our process, all with a different diameter. This is because the maximum allowable particle size varies through the process. For example, at the beginning, the particles must be less than 595 μ m, but once washed with ethanol, the allowable size increases to 4.76 mm. Each screen will be made of pharma-grade stainless steel and conform to Australian specifications, and all product that is not screened through will be discarded. This will

result in negligible product loss, mostly removing large crystalline impurities in the API, or impurities in purchased materials. All screens will be gravity-based.

3.6.3.2 Fitz Mill

Here, we select a Fitzpatrick D6Ax FitzMill. This is a standard Fitz Mill, as mentioned above in the process, for pharmaceutical applications. All materials are pharma-grade. The mill has a maximum throughput of 3,000 kg/h, but we plan to run the mill at 300 kg/h, meaning all material will move through the mill in 15 minutes for our batch size. The mill uses 5.6 kW of power.

3.6.3.3 V Mixers

Multiple times in the batch process, powders need to be mixed. We will use one mixer for all of the mixing processes in our tableting process, except for the mixing of povidone and ethanol, which occurs simultaneously with the main mixing operation. Both of our mixers will be V mixers, which are commonly used in powder-heavy pharmaceutical mixing to allow for diffusion mixing (Ability Fabricators, 2018). They will be made of stainless steel.

The main mixer will have a volume of 500 L. It will contain, at maximum, about 90 kg of powder and 8.3 kg of ethanol. It is important that the volume be much larger than the working volume to allow for proper mixing. Additionally, most specifications outline a maximum mixing weight, which is more important than the batch volume. Our smaller mixer, for 6 kg of povidone and 5 kg of ethanol, will be 100 L.

The mixers will be sourced from Sainty International Group, and the specifications are included below as Table 3.6.3.3-1.

Table 3.6.3.3-1. Mixer specifications.

Specification	Large mixer	Small mixer
Volume (L)	500	100
Maximum mixing weight (kg)	100	35
Rotation speed (rpm)	12	12
Power (kW)	2.2	1.1
Dimensions (m)	2.5x1.4x2.5	1.7x0.7x1.7

3.6.3.4 Dryer

Because the batches are aligned with the drying step earlier in the process, we can use the same dryer as earlier in the process. However, here we must dry for longer, as this is the last chance to remove liquid before the mixture moves to tableting.

The model of dryer for this process is the FZG-20 which can dry batches with a maximum drying capacity of 120 kg. It uses 48 trays for drying with a total drying surface area of 14.13 m². It will be operated at 35°C and 0.033 atm. The heating power of it is 21 kW which is provided by low-pressure steam. Here, it will be dried for 3 hours, to ensure that no ethanol or water is present, as this may damage the tablet press or cause degradation of the final product over time.

3.6.3.5 Tablet Press

Our tableting press will be a COMPRIMA 300 Tablet Press, from IMA Life. This tablet press can handle 300,000 tablets per hour, meaning our batches will take 30 minutes. It uses a press force of 82 kN, a speed of 138 rpm, and creates 5x16 mm tablets.

3.6.3.6 Bottling and Packaging

The bottling and packaging cost will not be designed at this time, but the price and specifications are estimated using available commercial estimates. The product will be bottled in 20 dram bottles, with 20 tablets per bottle.

3.6.4. Ethanol Use

In the tableting process, we use ethanol to facilitate the even distribution of API and excipients before the powder is compressed into tablets. Ethanol is also used earlier in the process. We remove ethanol through a batch drying process, then compress it before re-use. So, though there is a set amount of ethanol used in each batch, very little ethanol leaves the process.

Because the ethanol is being evaporated, it will be relatively pure, but impurities can accumulate if the ethanol is continually re-used. So, before re-use of ethanol, we will analyze the batch and test it for impurities, particularly impurities from earlier in the process. (We will also analyze for excipients added, but these will have a higher acceptance threshold because it is not harmful for these to accumulate, though large accumulations harm quality control.) If the batch does not meet criteria for reuse, it will be discarded.

We estimate that ethanol can be reused for, on average, ten batches. Each batch requires 8.3 kg of ethanol, which we can distribute across ten batches to estimate a per-batch input of 0.83 kg.

3.7 Ancillary Equipment

3.7.1 Chillers

In this process, a single chiller is employed to cool the ethylene glycol stream used for temperature control of the crystallizers. Based on the required cooling duty for the specified ethylene glycol flow rate, the chiller is designed to operate at approximately 40 kW. This capacity is sufficient to maintain the desired crystallization temperature and ensure consistent process performance.

3.7.2 Pumps

Multiple pumps are required throughout the process to ensure adequate material transport between unit operations. A positive displacement pump will supply water to the process operating at 313 W due to the larger volumetric flow rate.

Material transfer between the four cascading continuous stirred-tank reactors (CSTRs) is facilitated by three centrifugal pumps, each operating at 0.313 kW. Downstream of the reaction system, a centrifugal pump with a power requirement of 4.21 kW is used to transport the reacted stream to the reverse osmosis (RO) filtration unit.

Subsequent slurry transfer between crystallizers and to the centrifugation unit is accomplished using three centrifugal pumps, each with a power requirement of 0.022 kW. In addition, five centrifugal pumps are employed in the ethylene glycol cooling loop associated with the crystallization system, each operating at 0.128 kW to maintain the required coolant circulation rate.

Further pumping requirements arise from pH adjustment within the crystallizers. Three additional centrifugal pumps are used to deliver 30 wt% HCl solution and 25 wt% aqueous

ammonia. The pump supplying HCl to the first crystallizer operates at 0.129 kW, while the pumps delivering ammonia to the second and third crystallizer require 0.101 kW. The final ammonia feed to the third crystallizer has a significantly lower flow requirement and corresponding power consumption than the second.

Two pumps are required to facilitate the washing of the amoxicillin product. A centrifugal pump with a power requirement of 0.21 kW is used to deliver the 30 wt% ethanol solution to the washing unit. Additionally, a second centrifugal pump, operating at 0.21 kW, is employed to transfer the spent washing liquid from the washing unit to the wastewater treatment system.

Each pump calculation was found by calculating hydraulic power by using this equation:

$Power = dP * V$; $dP = P_{process} + P_{friction} + \rho gh$; where V is volumetric flow rate in (m³/s), Process is the pressure difference between source and destination, $P_{friction}$ is the frictional losses, and ρgh accounts for elevation changes

3.7.3 Tanks

Multiple storage tanks are required throughout the process to accommodate raw materials and support continuous operation. The 6-aminopenicillanic acid (6-APA) feed tank is designed with a volume of 12.25 m³, a diameter of 2.18 m, and a height of 3.27 m, providing approximately 90 days of storage capacity. Similarly, the hydroxyphenylglycine methyl ester (HPGME) feed tank has a volume of 11.25 m³, a diameter of 2.12 m, and a height of 3.18 m, also corresponding to 90 days of supply.

Due to the significantly higher volumetric demand, the water feed tank is substantially larger, with a volume of 763.2 m³, a diameter of 8.65 m, and a height of 12.98 m, providing

approximately 3 days of storage. Storage tanks for auxiliary chemicals are sized based on both process requirements and regulatory considerations. The tank for 30 wt% HCl solution has a volume of 1.5 m³, diameter 1.08 m, and height of 1.6 m, corresponding to approximately two weeks of storage (Safe Work Australia. (n.d.)). This capacity remains below regulatory threshold quantities, requiring only standard hazardous chemical management practices rather than additional placarding or manifest controls. Likewise, the 25 wt% aqueous ammonia tank has a volume of 2.2 m³, diameter 1.2 m, and height 1.9 m, providing approximately four weeks and similarly remains below regulatory threshold quantities, subject to the same standard handling requirements.

The ethylene glycol storage tank is designed with a volume of 22.55 m³, a diameter of 2.75 m, and a height of 4.13 m, corresponding to approximately 3 hours of inventory. Given that ethylene glycol is continuously recycled within the cooling loop, this capacity is sufficient to ensure uninterrupted operation while minimizing excess inventory.

The 30 wt% ethanol storage tank is designed with a volume of 75 m³, a diameter of 4 m, and a height of 6 m, corresponding to approximately 90 days of storage capacity. Intermediate product handling is achieved using two 100 L vessels, each with a diameter of 0.44 m and a height of 0.66 m, which are used to hold amoxicillin following the washing step and prior to drying. Each vessel is sized to accommodate a single batch of approximately 75 kg of amoxicillin.

Following drying, a larger storage tank with a volume of 2 m³, diameter 1.19 m, and height 1.79 m is used to hold the final amoxicillin product. This tank provides approximately 8 hours of storage capacity prior to transfer to the tableting process, allowing for operational flexibility and continuity between downstream processing steps.

All storage tanks were designed with an allowance of 20% headspace to accommodate vapor expansion and operational flexibility. Tank dimensions were determined assuming a height-to-diameter ratio of 1.5, consistent with standard design practices for vertical cylindrical vessels.

3.7.4 Pressure Valves

Following RO filtration, the process stream pressure must be reduced from 10 bar to approximately 1 atm prior to crystallization. This pressure reduction is achieved using two pressure control valves arranged in series, with the first reducing pressure from 10 bar to 5 bar and the second from 5 bar to near-atmospheric conditions. Through these valves, 1145 kg/h of water flows, and this will be the basis for the pressure valve diameter. A piping diameter can be calculated assuming 1 m/s of piping velocity, giving a piping diameter of 2 cm. So, the pressure valves must have a diameter of 2 cm.

3.7.5 Feeders

At the beginning of the process, we have to feed solids at an exact ratio. For this, we select two rotary airlock feeders (one for 6-APA, one for HPGME). Assuming a bulk density of 500 kg/m³ for both components, and a fill efficiency of 0.5, which is standard, we can calculate the feeder size:

$$V_p = \frac{Q_v}{N\eta} \text{ (Eq. 3.7.5-1)}$$

where V_p is volume per revolution, Q is volumetric flow of solid, and n is efficiency. So, given our initial flows of 7.6 kg/h of 6-APA and 6.5 kg/h of HPGME, and running at 20 RPM, we get a capacity of 25 mL per revolution for 6-APA, and 22 mL per revolution for HPGME.

3.7.6 Condensers

The evaporated liquid from the vacuum tray driers in the drying stage and the tableting stage needs to be condensed prior to waste water treatment and recycling respectively. We designed two condensers to accomplish this. Both condensers use 10°C cooling water as the heat sink which will exit the condenser at 30°C. The required energy removal is the latent heat of vaporization per batch: 43700 kJ and 7530 kJ from the drying stage and tableting stage respectively. The flow rate of cooling water needed was calculated by equation 3.7.6-1:

$$Q = mc_p(T_{out} - T_{in}) \text{ (Eq. 3.7.6-1)}$$

where c_p is the isobaric heat capacity of the cooling water, $4180 \frac{J}{kg K}$ and m is the mass flow rate of cooling water. From this, we found the drying and tableting stages required 521 kg and 98.7 kg of 10°C cooling water per batch respectively. The pumps for the cooling water were designed utilizing the same methodology as those computed in section 3.7.2 to yield required power usage of 28.3W for drying and 1.00W for tableting.

4. Recommended Design

4.1 Process Flow Diagram

The equipment convention used in this Process Flow Diagram was developed in accordance with common industry practices for process documentation (*Process Flow Diagram - Processdesign*, 2015).

4.1.1 Equipment Designations

Equipment is identified using standardized letter abbreviations, as summarized below:

- B - Conveyor Belt
- E - Exchanger
- F - Filtration
- FD - Feeder
- P - Pump
- R - Reactor
- S - Separation Unit
- CF - Centrifuge
- PV - Pressure Control Valve
- DR - Dryer
- ML - Fitz Mill
- MX - Mixer
- PO - Wastewater Treatment Plant
- PS - Tablet Press
- TK - Tank

4.1.2 Section-Based Numbering System

Each process section is assigned a numerical range, as summarized below. Equipment numbering begins at X01 within each area, where the leading digit corresponds to the process section and the trailing digits denote the specific unit within that section. Full equipment labels are formatted using the standardized letter abbreviation followed by the assigned area number (e.g., R-101, CF-301).

The numerical ranges for each process section are summarized below:

Reactor Section: 100s

Reverse Osmosis Section: 200s

Crystallization and Centrifugation Section: 300s

Washing and Drying Section: 400s

Tableting Section: 500s

Process streams labeling aligns with its corresponding process area. Stream numbers begin in the 1000s and follow the same section-based numbering convention. The jacketed system in the crystallization and centrifugation section is distinguished using subunit numbers in the 3100 series.

For ease of reading, the process flow diagram is divided by section.

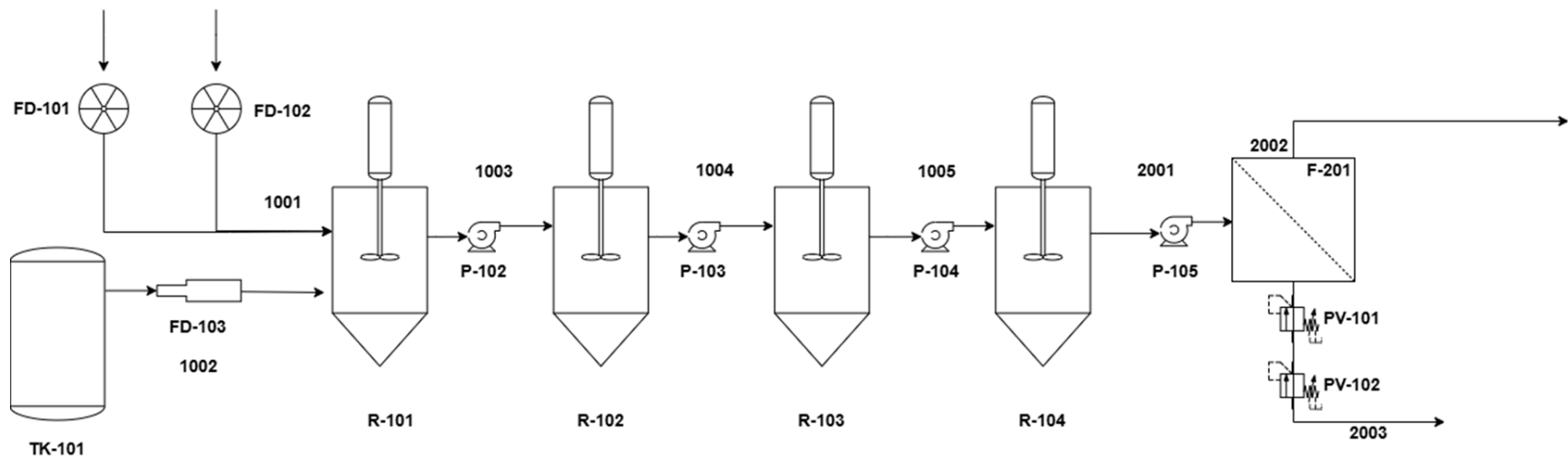


Figure 4.1-1. Reactor and reverse osmosis process flow diagram.

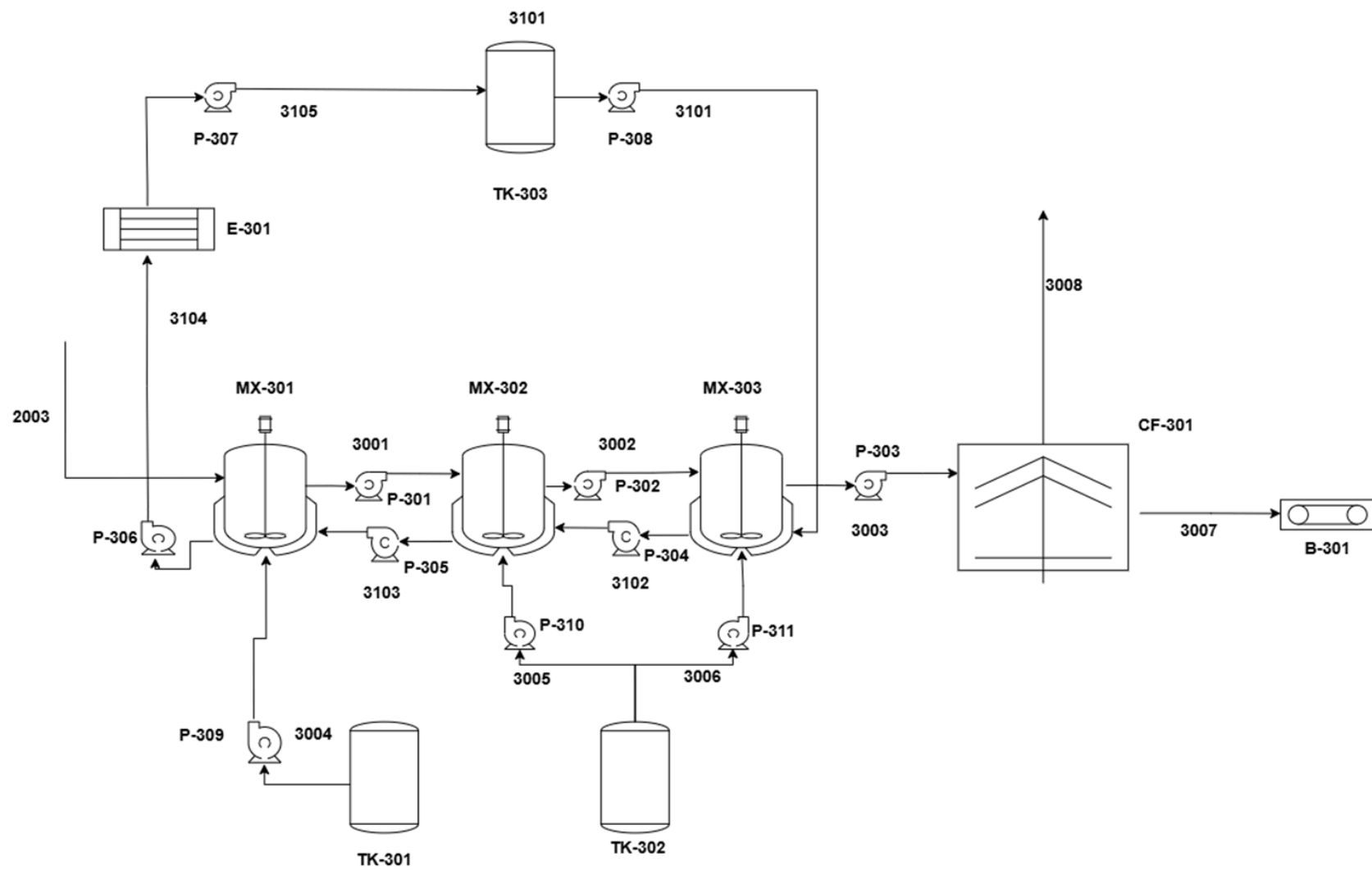


Figure 4.1-2. Crystallization and centrifugation process flow diagram.

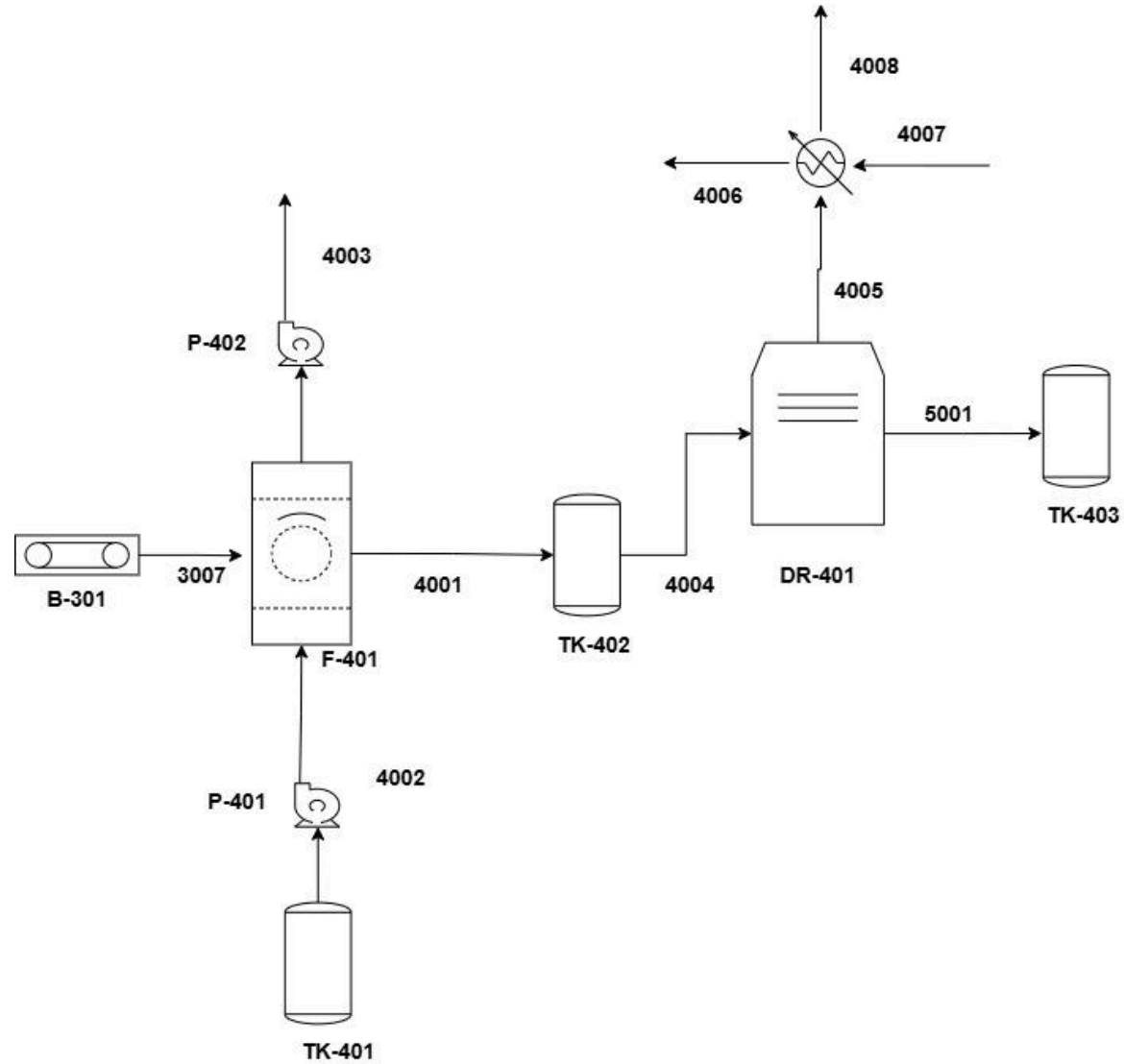


Figure 4.1-3. Washing and drying process flow diagram.

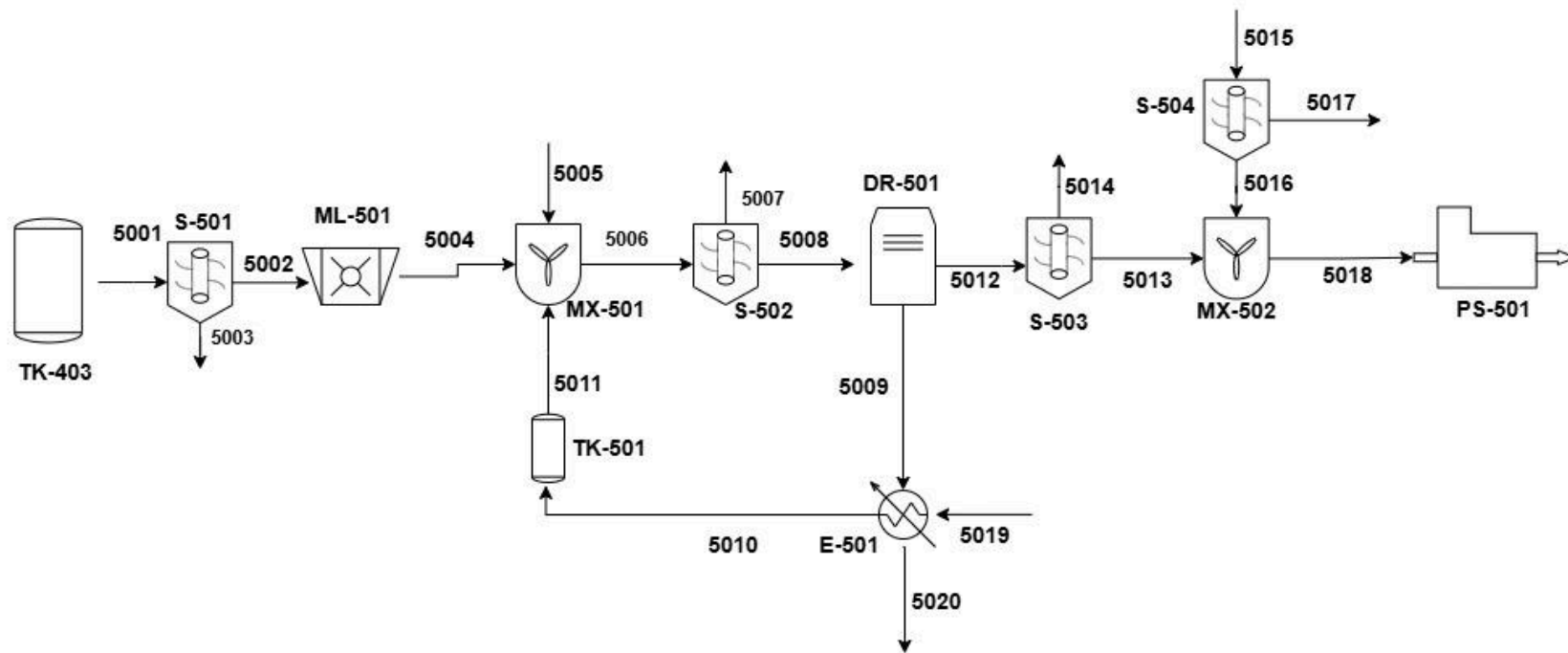


Figure 4.1-4. Tableting process flow diagram.

4.2 Material Balances

Below, we include stream tables of our material balances. The first two are for the continuous portion of our process, where each component carries units of kg/h. They are separated for readability. The third is for the batch portion of the process, where each component carries units of kg/batch. Temperature, pressure and pH is reported when necessary to the process (i.e. pressure in reverse osmosis, pH in crystallization).

Table 4.2-1. Reactor and reverse osmosis stream table. Flow rates are reported in kg/h.

	1001	1002	1003	1004	2001	2002	2003
6-APA	7.6	1.14	0.46	0.23	0.15	0.0001	0.15
HPGME	6.5	0.975	0.39	0.2	0.13	0.0006	0.13
Water	10600	10600	10600	10600	10600	9455	1145
Amoxicillin	-	12.7	14	14.5	14.6	0.07	14.5
MeOH	-	1	1.1	1.1	1.1	0.99	0.11
Total	10614	10616	10616	10616	10616	9456	1160
Pressure (atm)	1	1	1	1	1	10.87	10.87
Temperature (°C)	25	25	25	25	25	25	25

Table 4.2-2. Crystallization, centrifugation, and washing stream table. Flow rates are reported in kg/h.

Stream #	3001	3002	3003	3004	3005	3006	3007	3008	3101	3102	3103	3104	3105	4001	4002	4003
6-APA	0.15	0.15	0.15	-	-	-	3e-4	0.150	-	-	-	-	-	-	-	0.0003
HPGME	0.13	0.13	0.13	-	-	-	3e-4	0.140	-	-	-	-	-	-	-	0.0003
Water	1148	1150	1149	3.08	1.84	0.01	3.58	1453	-	-	-	-	-	2.98	5.01	5.61
Amoxicillin	14.5	14.5	14.5	-	-	-	11.97	2.53	-	-	-	-	-	11.95	-	0.02
MeOH	0.11	0.11	0.11	-	-	-	3e-4	0.110	-	-	-	-	-	-	-	0.0003
HCl	1.32	1.32	1.32	1.32	-	-	0.004	1.666	-	-	-	-	-	-	-	0.004
EtOH	-	-	-	-	-	-	-	-	-	-	-	-	-	0.72	2.15	1.45
NH3	-	0.61	0.62	-	0.61	0.01	0.002	0.788	-	-	-	-	-	-	-	0.002
EG soln.	-	-	-	-	-	-	-	-	7570	7570	7570	7570	7570	-	-	-
Total	1164	1167	1166	4.4	2.45	0.02	15.56	1458	7570	7570	7570	7570	7570	15.65	7.16	7.09
Temperature (°C)									-3	2	2.7	7.8	-3			
pH	1.5	3.5	4.7	4.7	4.7	4.7	4.7	4.7	9	9	9	9	9	4.7		

Table 4.2-3. Drying and tableting stream table. Flowrates are reported in *kg/batch*.

	4004	4005	4006	4007	4008	5001	5002	5003	5004	5005	5006	5007	5008	5009	5010	5011	5012	5013	5014	5015	5016	5017	5018	5019	5020
Amoxicill in	75	0.02	0.02	-	-	75	75	~ 0	75	-	75	~ 0	75	0	0	-	75	75	~ 0	-		-	75	-	-
EtOH	-	-	-	-	-	-	-	-	-	0 or 8.3 ¹	8.3	0	8.3	8.3	8.3	8.3 or 0 ¹	-	-	-	-		-	-	-	-
Water	18.1	18.1	18.1	521	521	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		-	-	89.7	-
Cellulose	-	-	-	-	-	-	-	-	-	8.55	8.55	~ 0	8.55	0	0	-	8.55	8.55	~ 0	-		-	8.55	-	-
Mg stearate	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1.05	1.05	~ 0	1.05	-	-
Povidone	-	-	-	-	-	-	-	-	-	6	6	~ 0	6	0	0	-	6	6	-	-	-	-	6	-	-
Total	93.1	18.1 2	18.1 2	521	521	75	75	0	75	14.5 5	97.8 5	0	97.8 5	8.3	8.3	8.3	89.5 5	89.5 5	0	1.05	1.05	0	90.6	89.7	89.7

¹ In most batches, ethanol will be recycled, but in some, it will not. See section 3.6.4 for a discussion on ethanol use in the tableting process.

4.3 Reactors

The process begins by feeding 7.6 kg/h of 6-APA and 6.5 kg/h of HPGME into two rotary airlock feeders that meter each reactant into the CSTR. Due to the solid crystalline structure of both 6-APA and HPGME, a dry feeding system was used to allow for continuous operation and maintain control over reactant stoichiometry. Water will be stored in a 763.2 m³ storage tank (TK-101, 8.65 m in diameter and 12.98 m in height), which holds 3 days of supply.

Each of the four reactors are identical. Each reactor has a tank diameter of 0.38 m and a height of 0.56 m, for a total volume of 62.2 L. Each reactor has a residence time of 1.38 minutes. Each reactor has a 45° pitched-blade turbine, with a diameter of 0.11 m. It will be operated at 380 rpm and require 11.7 kW of power. Four vertical baffles will be included in each reactor, with a width of 3.8 cm and a thickness of 0.4 cm. The reactors contain penicillin G acylase (PGA) immobilized on agarose beads at an activity concentration of 50,000 IU/cm³. Agitation provided by the pitched-blade turbine maintains the agarose beads in suspension to promote adequate mixing and mass transfer throughout the reactor. Internal retention screens are included to prevent loss of the immobilized enzyme beads between reactors while allowing the liquid stream to flow continuously through the reactor cascade. The immobilized enzyme is replaced every two weeks during scheduled plant downtime. In the reactors, 6-APA and HPGME will react to produce amoxicillin and methanol. The reactor cascade will reach 98% conversion of 6-APA into amoxicillin.

In between each reactor will be a centrifugal pump (P-102, P-103, and P-104), for a total of 3 pumps. Each will have a power of 313 W.

Leaving the reactor will be 14.6 kg/h of amoxicillin, 0.15 kg/h of 6-APA, 0.13 kg/h of HPGME, 10,600 kg/h of water, and 1.1 kg/h of methanol (in stream 2001). This will flow into a

centrifugal pump (P-105), which will add 10 bar of pressure to the stream before the reverse osmosis unit and use 4.21 kW of power.

4.4 Reverse Osmosis

Then the stream with product (stream 2001) flows into a reverse osmosis membrane (RO-201) to remove most of the water. Water is removed here to make downstream processes more efficient. The membrane has a salt rejection rate of 99.5%, and will have an area of 6.5 m². Almost all of the amoxicillin, 6-APA and HPGME is retained, and 86% of the water and methanol permeates, with the remaining water being retained. For a material balance on the reverse osmosis process, see table 3.2-2. The permeate (stream 2002) is sent to wastewater treatment, while the retentate (stream 2003) is sent forward in the process to crystallization.

Two pressure control valves will be used to bring the pressure down; the first (PV-101) will release the pressure to 5 bar gauge, and the second (PV-102) will release the pressure to 0 bar gauge, or atmospheric pressure. These will be 2 cm in diameter.

4.5 Crystallization

From reverse osmosis, the stream with product (stream 2003) flows into a cascade of three mixed-suspension mixed-product removal (MSMPR) crystallizers (MX-301, MX-302 and MX-303). The three crystallizers are identical in construction, but are run at different conditions. Each crystallizer has a diameter of 1.36 m and a height of 2.03 m, for a total volume of 2.93 m³. Each crystallizer has a 45° pitched-blade turbine, with a diameter of 0.45 m. It will be operated at 90 rpm and require 18.5 kW of power. Between the crystallizers will be two identical centrifugal pumps (P-301 and P-302), each with a power of 22 W.

Each of the three crystallizers operates at a different pH to crystallize selectively. Along with the product stream exiting reverse osmosis (stream 2003), 4.4 kg/h of 30 wt% aqueous hydrochloric acid (in stream 3004 from tank TK-301) enters the first crystallizer (MX-301) at pH 1.5, well below the isoelectric point of amoxicillin. Then, all of the contents from MX-301 (in stream 3001) enter the second crystallizer (MX-302), along with 2.45 kg of 25 wt% aqueous ammonia (in stream 3005 from tank TK-302) at a pH of 3.5. Then, all of the contents from MX-302 (in stream 3002) enter the third crystallizer (MX-303), along with 0.02 kg of additional 25 wt% aqueous ammonia (in stream 3006 from tank TK-302). This gives the third crystallizer a pH of 4.7, the isoelectric point of amoxicillin.

HCl will be stored in a 1.5 m³ storage tank (TK-301, 1.08 m in diameter and 1.6 m in height). This tank holds 2 weeks of storage in accordance with regulatory standards. It will be pumped (as stream 3004) by a centrifugal pump (P-309), operating at 70 mW.

Aqueous ammonia will be stored in a 2.2 m³ storage tank (TK-302, 1.2 m in diameter and 1.9 m in height). This tank holds 4 weeks of storage in accordance with regulatory standards.

Stream 3005 will be pumped by a centrifugal pump (P-310) operating at 50 mW, while stream 3006 will be pumped by a centrifugal pump (P-311) operating at 0.4 mW.

4.5.1 Cooling of the Crystallizers

The three crystallizers are cooled by ethylene glycol, which flows counter-current through the MSMPR cascade. 7,570 kg/h of ethylene glycol mixture enters MX-303 at -3°C (in stream 3101), then flows through MX-302 and MX-301 exiting at 7.8°C (in stream 3104). Below is a table of the cooling conditions of the three crystallizers.

Table 4.5-1. MSMPR Specifications.

MSMPR	pH	Operating Temp. (°C)	EG Entering Temp. (°C)	EG Exiting Temp. (°C)	Heat Removal (kW)
MX-301	1.5	10	2.7	7.8	26
MX-302	3.5	8	2	2.7	3.4
MX-303	4.7	5	-3	2	5.1

The ethylene glycol then cycles through a chiller (E-301) operating at 44 kW, and then it is returned to the storage tank (TK-303, 22.55 m³ in volume, diameter of 2.75 m, height of 4.13 m). Between each of the 5 units (MX-301, MX-302, MX-303, E-301 and TK-303) is a centrifugal pump (P-304, P-305, P-306, P-307, and P-308) for the ethylene glycol, each using a power of 128 W.

4.6 Centrifugation

The product then flows into a disc stack centrifuge (CF-301) using a 22W centrifugal pump (P-203). The DSC has 50 discs at 45°, with an inner radius of 25 cm and an outer radius of 45 cm. It will rotate at 600 rpm and consume 130 W of power. The purpose of the centrifuge is to remove the bulk of the water remaining in the process, along with the remaining starting materials, 6-APA and HPGME.

11.97 kg/h of amoxicillin will be sent to washing, with 3.58 kg/h of water and minimal amounts of other components (in stream 3007). 1,453 kg/h of water, 2.53 kg/h of amoxicillin, and most of the remaining 6-APA and HPGME will be sent to wastewater treatment (in stream 3008).

4.7 Washing and Drying

4.7.1 Washing

The wetcake (stream 3007) will be moved via a conveyor belt (B-301) to a rotary drum filter F-401). The rotary drum filter will use a filter area of 0.5 m^2 and rotate at a rate of 0.1 rpm, for a residence washing time of 6 minutes. Using a 2:1 length to diameter ratio, the rotary drum length will be 1.01 m and the diameter will be 0.52 m. The filter is designed for a cake thickness of 5 mm, consistent with industry standard to prevent cake cracking (Rotary Vacuum Drum Filter, n.d.). The purpose of the washing step is to ensure the last of the 6-APA and HPGME is removed before the API is tableted. This is done based on differences in solubility in water and ethanol.

To it, 7.16 kg/h of 30% ethanol (stream 4002) will be added from a storage tank (TK-401). TK-401 is a 75 m^3 storage tank (4 m in diameter and 6 m in height), holding 90 days of supply. Flowing out will be an ethanol-rich stream with the remainder of the 6-APA and HPGME (stream 4003), and a water-rich stream with the amoxicillin (stream 4001). This will be sent to a storage tank (TK-402), where the process will be converted from continuous to batch. TK-402 is a 100 L storage tank (0.44 m in diameter and 0.66 m in height), holding one batch of amoxicillin. TK-402 is only represented once, but in reality there will be two in case of process deviations.

4.7.2 Drying

Once per batch, an operator will move product from the storage tank (TK-402) to the dryer (DR-401). This batch will be 98.2 kg total, and contain 75 kg amoxicillin, 4.5 kg ethanol, and 18.7 kg water. The dryer is a vacuum tray dryer, which has a total drying capacity of 120 kg

spread across 48 trays. The total drying surface area is 14.13 m². It will be operated at 35°C and 0.033 atm for 37 minutes. The heating power of it is 21 kW which is provided by low-pressure steam. The amoxicillin will be transferred into a 2 m³ storage tank (TK-403, 1.19 m in diameter and 1.79 m in height) and quality tested (see the discussion in Section 6.2).

4.8 Tableting

For each batch, 75 kg of API will be used to create tablets of 500 mg each. First, the amoxicillin API (stream 5001) is passed through a 595- μ m aperture screen (S-501) to remove large particles and a Fitz mill (ML-501). The Fitz Mill is a Fitzpatrick D6Ax FitzMill, which will be run at 300 kg/h, such that material will exit 15 minutes after it enters. The mill will use 5.6 kW of power, but only when active.

Then, 8.55 kg of cellulose (stream 5005) is added with the amoxicillin to a V mixer (MX-501), and it is mixed for 30 minutes. This V mixer is 500 L, and has dimensions of 2.5x1.4x2.5 m. It will rotate at 12 rpm, and it uses 2.2 kW of power when active. Then, to the same V mixer (MX-501), we add 3.3 kg of ethanol (also represented as stream 5005), and mix for 15 additional minutes.

In a separate vessel (TK-501), we will add the balance of ethanol (5 kg) and 6 kg of povidone. The povidone will be dissolved in the ethanol, and this mixture (stream 5011) will be added to the V mixer (MX-501) and be mixed for 15 minutes.

The total of amoxicillin, ethanol, cellulose, and povidone (stream 5006) will then pass through a 4.76-mm aperture screen (S-502) to remove large particles and be transferred to a dryer (DR-501). This dryer will be the exact same as the earlier drying process (DR-401, see section 4.7.2). It will dry for 3 hours under vacuum, using 21 kW while active.

The ethanol (stream 5009) will pass through a condenser (E-501) and be quality tested to determine the level of impurities. If the impurities are sufficiently low, the ethanol will be recycled for the consecutive batch. If not, to prevent accumulation, it will be discarded and a fresh batch of ethanol will be used. We estimate that each batch of ethanol will be able to be used 10 times before impurities accumulate (see section 3.6.4).

The solids from the dryer (stream 5012) will then be passed through a 1.2-mm aperture screen (S-503) to remove large particles. Then, half of the exiting quantity (half of stream 5013) will be added to a smaller V mixer (MX-502). This V mixer is 100 L, and has dimensions of 1.7x0.7x1.7 m. It will rotate at 12 rpm, and it uses 1.1 kW of power when active. To it, 1.05 kg of magnesium stearate (stream 5015) will be passed through a 500- μ m aperture screen (S-504) and added to the V mixer (as stream 5016). It will be mixed for 20 minutes, then the remainder of stream 5013 is added, and mixed for another 10 minutes.

Leaving the V mixer, stream 5018 will be sent through a tablet press (PS-501). The tablet press is a COMPRIMA 300 Tablet Press. This tablet press can handle 300,000 tablets per hour, meaning our batches will take 30 minutes. It uses a press force of 82 kN, a speed of 138 rpm, and creates 5x16 mm tablets. Then, the tablets (stream 5014) will proceed to bottling.

5. Economics

This process will take place in Australia, but the report is being written in the United States. Therefore, we will need to use two currencies: United States dollars (USD) and Australian dollars (AUD). Most equipment costs will be in USD, but costs specific to Australia will use AUD. For this report, we will use the conversion rate of \$1.00 USD = \$1.43 AUD, as of March 2026.

For this economic assessment, we assume a 1-year construction period, with 20 years of operation. We assume that the credits for working capital at the end of the process will be approximately equal to our shutdown costs, meaning that we do not need to account for a shutdown cost or credit.

5.1 Revenue Estimation

In Australia, the price of medicines is determined by the Pharmaceutical Benefits Scheme (PBS), an agency of the government. They set an approved ex-manufacturer price (AEMP), which is the price that they will pay the manufacturer for a packaged drug product. For amoxicillin, this value is \$2.50 AUD (\$1.75 USD) for a pack of 20 tablets of 500 mg each, as of November 2025. At a total production target of 160,000,000 tablets per year, or 8,000,000 packages, this would yield a revenue of \$20 million AUD (\$14.0 million USD) per year.

5.2 Capital Costs

5.2.1 Equipment Costs

To find the purchase cost of most equipment, we used correlations from *Rules of Thumb in Engineering Practice* (Woods, 2007), Appendix D. Where our values fell outside of the

applicable ranges, we chose the low value of the range as an upper bound for the equipment cost. This only accounts for the cost of the equipment itself, not adjusting for installation. These values are listed in Table 5.1.1-1, reported in U.S. Dollars.

Table 5.2.1-1. Purchase Cost of Equipment.

Equipment Number	Equipment	Cost (USD)
R-101 - R-104	CSTR	\$62,900
F-201	RO	\$18,400
MX-301 - MX-303	Crystallizers	\$794,000
CF-301	DSC	\$476,000
F-401	Rotary drum filter	\$123,000
DR-401, DR-501	Dryer	\$123,000
N/A	Tempered glycol system	\$75,100
FD-101	Rotary Feeders	\$5,100
FD-102		\$5,100
B-301	Conveyor belt	\$6,100
MX-501	V Mixers	\$54,700
MX-502		\$54,700
E-401	Condensers	\$44,800
E-501		\$44,800
ML-501	Mill	\$23,700
S-501 - S-504	Screens	\$188,000
PS-501	Tablet Press	\$71,600
FD-103	Water feed pump	\$12,300
P-102	Pumps	\$12,300

P-103	(including spares)	\$12,300
P-104		\$18,700
P-105		\$12,300
P-201		\$12,300
P-202		\$12,300
P-203		\$12,300
P-209		\$12,300
P-210		\$12,300
P-211		\$12,300
P-401		\$12,300
P-402		\$627,000
TK-101	Water tank	\$36,400
TK-301	Small tanks	\$67,000
TK-302		\$69,000
TK-303		\$102,000
TK-401		\$36,000
TK-402		\$36,000
TK-403		\$36,000
TK-501		\$23,000
PV-101, PV-102	Pressure valves	\$70,000
N/A	Bottling line	\$70,000.00

5.2.2 Other Capital Costs

To account for other costs associated with equipment such as construction, installation, initial working capital, and engineering, the equipment cost is multiplied by a Lang factor.

Towler and Sinnott propose a Lang factor of 4.3 for solids and fluids processing. This gives a fixed capital investment of \$14.75 million USD.

Land costs, which is not accounted for in the Lang factor can be estimated as 8% of the equipment costs for the process. For our process, this would be 8% of \$3.43 million USD, or \$270,000 USD.

Working capital was estimated as 15% of the fixed capital investment for the process. For our process, this would be 15% of \$14.75 million USD, or \$2.2 million USD.

This gives a total capital investment of \$17.2 million USD.

5.3 Income Tax and Depreciation

Australia operates on a two-tiered income tax system. Most businesses are taxed at 30%, however, businesses with annual turnover (income) below \$50 million AUD to use a lower tax rate of 25%. Our plant qualifies for this rate based on our income, as stated above.

Additionally, we can depreciate our capital investment against our income tax requirements. We choose to use a 7-year flat-line depreciation, meaning that for the first 7 years of plant life, we subtract one-seventh of the capital cost from our profit to get our taxable income. For the remaining years, our taxable income is our profit.

5.4 Operating Costs

5.4.1 Variable Operating Costs

5.4.1.1 Raw Materials

Our process contains a number of raw materials that must be purchased from external vendors. This includes our initial reactants, but also solvents and excipients in the tableting process. The raw materials are priced with a basis of 7200 hours per year of production (24 hours per day, 300 days per year). The cost of raw materials per year is included below as Table 5.2.1-1, and a full calculation of raw material prices, including sources, is included in Appendix 10.4.

Table 5.2.1-1. Raw Material Costs per Year.

Material	Cost per Year (USD)
6-APA	\$2,001,000.00
HPGME	\$518,000.00
HCl	\$2,000.00
EtOH	\$15,000.00
NH3	\$2,000.00
EG	\$14,000.00
Cellulose	\$54,000.00
Mg stearate	\$16,000.00
Povidone	\$83,000.00
Bottles and lids	\$1,440,000.00
Total	\$4,145,000.00

5.4.1.2 Labor

Labor requirements were estimated using standard correlations based on process equipment, based on typical workers per unit per shift. Each major unit operation in the process was assigned a labor requirement (Peters et al., 2003). A total of 10 operators per shift is needed across all units. For continuous operation, a 5-shift rotation schedule was implemented, resulting in a total workforce of 50 operators. The average operator salary was estimated based on industrial wage data for manufacturing workers in Sydney, with an annual salary of \$68,900 AUD/year (*Factory Worker Salary in Sydney NSW*, 2026). An additional 15% of operating labor was added for both operating supervision and payroll overhead, resulting in a total labor cost of approximately \$4.5 million AUD/year.

5.4.1.3 Utilities

The plant utilities were categorized into electricity, cooling water, process water, steam, and wastewater treatment.

Electricity consumption was calculated for all major and ancillary equipment, including pumps, mixers, reactors, and crystallizers. For continuous operations, annual electricity usage was determined using the amount of power over 7200 hours of operation per year, while batch processes were calculated based on power over a single batch duration for 900 batches per year. The cost of electricity for each unit was determined from the total electricity consumption (kWh/year) and the industrial flat electricity rate in New South Wales, taken as \$0.2349 AUD/kWh (*Compare Commercial Electricity Prices - Electricity Brokers*, 2025).

The largest contributors to electricity usage were the CSTR impellers, MSMR crystallizer impellers, and the ethylene glycol recirculation and chilling system

Cooling and process water estimates were determined using a unit price of \$0.35 AUD/kL based on data from the Australian Bureau of Statistics (*Water Account, Australia, 2023-24 Financial Year* | Australian Bureau of Statistics, 2025). Steam demand required for the drying operations was converted to energy units (GJ/year) and priced using industrial rates in Australia. Based on data from the Australian Competition & Consumer Commission, steam costs were estimated at \$15 AUD/GJ (*Tight Supply Forecast for Australia's East Coast Gas Market in Q2 2026* | ACCC, 2025.). Further details on the cost estimation of wastewater treatment are provided in Section 6.3.1. In total, plant utilities amounted to \$572,000 AUD annually.

5.4.2 Fixed Operating Costs

Insurance and maintenance costs for the facility were estimated based on a total equipment cost of \$3.43 million USD. Insurance was assumed to be 0.7% of the total equipment cost, resulting in an annual insurance expense of approximately \$24,000 USD (Peters et al., 2003). Maintenance costs were estimated at 5% of the total equipment cost, corresponding to an annual expense of approximately \$170,000 USD (Peters et al., 2003). Property taxes were not included in the fixed operating costs. In New South Wales, property tax is only applied when land value exceeds a threshold of \$1,075,000 AUD (CleatTax Accountant, 2025). Since the estimated land value for this facility is approximately \$390,000 AUD, no property tax is incurred.

5.4.3 Total Cost of Production

Our total cost of production is calculated as the sum of our fixed and variable operating costs. A summary is included below as Table 5.4.3-1.

Table 5.4.3-1. Production Costs Summary.

Category	Price (USD)
Property tax	\$0
Insurance cost	\$24,000
Maintenance	\$170,000
Labor	\$3,130,000
Fixed Total	\$3,320,000
Raw Materials	\$4,150,000
Waste Management	\$180,000
Utilities	\$400,000
Variable Total	\$4,730,000
Cash cost of Production	\$8,050,000
Selling/Marketing	\$400,000
Total Cost of Production	\$8,450,000

5.5 Indicators of Success

To evaluate the financial viability of the proposed amoxicillin production facility, a comprehensive economic analysis was conducted. Key financial performance indicators, including Net Present Value (NPV), Return on Investment (ROI), and Internal Rate of Return (IRR), were used to assess the profitability and attractiveness of the project.

5.5.1 Net Present Value

Net Present Value (NPV) represents the present value of projected future cash flows discounted to account for the time value of money. A discount rate (r) of 10% was selected, consistent with the weighted average cost of capital (WACC) and typical return expectations for pharmaceutical industry investments. NPV was calculated using this equation:

$$NPV = \sum \frac{C_t}{(1+r)^t} - C_0$$
 where C_t is the net cash flow in year t and C_0 is the initial capital investment.

The calculated NPV for the project is \$20.7 million USD, indicating that the process is expected to generate significant value over its lifetime and is financially viable.

5.5.2 Return on Investment

Return on Investment (ROI) measures the profitability of the project relative to the initial capital investment. It was calculated using this equation:

$$ROI = \frac{Net\ Return - Initial\ Investment}{Initial\ Investment} \times 100$$

The ROI for the proposed facility was calculated to be 404%, demonstrating a substantial return relative to the initial investment and further supporting the economic feasibility of the design.

5.5.3 Internal Rate of Return

The Internal Rate of Return (IRR) is the discount rate at which the NPV of the project equals 0, and it serves as a key indicator of investment attractiveness. The IRR was determined using the IRR function in Microsoft Excel based on the projected cash flows. The calculated IRR is 26%, which exceeds the assumed discount rate range of 10–15% typically required for pharmaceutical investments. This result indicates that the project is highly attractive from an investment perspective.

6. External Effects

6.1 Safety

6.1.1 Chemical Compatibility

The reactivity matrix was created using CAMEO Chemicals. The chemicals within this matrix are stored within the same facility, so understanding the risks associated with unintended contact between components is critical to ensure proper safety precautions are put into place. Some of these chemicals will come into contact with each other as part of the process such as the 25% ammonia solution and water, so understanding any associated risks with those interactions is critical to ensure a design mitigates risk. The reactivity matrix can be found in Figure 6.1.1-1.

The 25% ammonia and 30% HCl solutions are incompatible with each other and upon contact will generate gas and heat. The reaction between these two solutions can be violent or explosive, therefore ensuring complete separation between them is necessary to avoid unintended contact. This is the only fully incompatible mixture in our process.

Mixing HCl with ethanol, methanol, or EG all generate heat, but do not pose any other risks. Very small amounts of HCl will come into contact with ethanol during the washing process, but due to the dilution of HCl in the mixture, the temperature change is negligible.

Mixing 25% ammonia with water generates heat but poses no other risks.

Mixing HCl with water creates a corrosive solution and generates heat and gas, but poses no other risks.

	WATER				
ETHANOL	Compatible 	ETHANOL			
AMMONIA, SOLUTION, WITH MORE THAN 10% BUT NOT MORE THAN 35% AMMONIA	Caution  Generates heat	Compatible 	AMMONIA, SOLUTION, WITH MORE THAN 10% BUT NOT MORE THAN 35% AMMONIA		
METHANOL	Compatible 	Compatible 	Compatible 	METHANOL	
ETHYLENE GLYCOL	Compatible 	Compatible 	Compatible 	Compatible 	ETHYLENE GLYCOL
HYDROCHLORIC ACID, SOLUTION	Caution  Corrosive Generates gas Generates heat	Caution  Generates heat	Incompatible  Corrosive Generates gas Generates heat Intense or explosive reaction	Caution  Generates heat	Caution  Generates heat

Figure 6.1.1-1 Reactivity Matrix

6.1.2 Chemical and Physical Hazards

There are several physical and chemical hazards associated with this process that can be dangerous to both employees and those in the areas surrounding the plant. Credible events were identified for each section of the process.

The reaction to form amoxicillin produces methanol, which is acutely toxic through ingestion, inhalation, and touch. A pipe rupture between reactors R-103 and R-104 would release a stream containing 1.1 kg/h of methanol. This would pose a danger to workers in the area through inhalation and dermatological contact. However, the methanol is diluted with 10600 kg/h of water, greatly reducing the impact of the methanol release.

The reverse osmosis step does not pose any significant safety events to the workers or the surrounding areas. The maximum pressure exhibited during RO is 10 bar which will not pose a significant threat.

The crystallization process utilizes both 25% ammonia in water and 30% HCl in water, both of which are corrosive liquids. The failure of the pump seal on (HCl pump) would release upwards of 4.4 kg/h of 30% HCl as a liquid. The failure of the pump seal on (pump for 3005)

would release upwards of 2.5 kg/h of 25% ammonia as a liquid. Both of these scenarios would endanger the workers as both would accumulate as corrosive puddles on the plant floor, allowing for accidental contact with them. Due to the liquid nature of both of these spills, the consequences would be confined to the plant itself and not the surrounding areas.

The washing process utilizes a 30% ethanol washing liquid stream, which is a flammable liquid. A pump seal failure of pump (stream 4002) would release upwards of 7.2 kg/h of 30% ethanol. This would create the possibility of a pool fire and endanger workers.

The tableting process contains a milling step in which the amoxicillin trihydrate crystals are ground to sizes less than 500 μm in diameter. When mixed with air, they form a dust-air mixture that is capable of creating a dust explosion. The effects of a dust explosion would likely extend to beyond the plant and cause damage to surrounding buildings.

6.1.3 Safe Work Practices

The equipment itself can pose a risk of injury to operators, technicians, and maintenance workers. Safe work practices will be implemented to prevent injury or death to workers when interacting the equipment.

One practice to implement are lockout-tagout (LOTO) procedures on equipment to ensure they are shut down, de-energized, and locked to prevent accidental start up during maintenance. Equipment such as the centrifuges with rapidly moving parts and are high energy must be fully de-energized and shutdown prior to maintenance, so LOTO procedures would prevent energy from being re-introduced and creating a dangerous situation for those performing maintenance on the centrifuge.

During maintenance and cleaning, personnel may need to access the inside of equipment such as storage tanks, centrifuges, and crystallizers. During these instances, confined space entry permits will be needed if a person is required to physically enter that equipment. These permits will allow for the assessment of hazards within the confined space, ensure oxygen concentrations are within an acceptable range, and ensure all safety measures are in place prior to entry.

Proper personal protective equipment (PPE) for all workers is necessary to prevent injury to workers and to protect product purity. The facility will need to remain sterile, so gowning for all employees on the manufacturing floor is mandatory to prevent contamination from outside sources. The gowning will include sterile coveralls referred to as a “bunny suit” as well as gloves and a face mask.

6.1.4 Maximum Credible Event

The maximum credible event for this process is a dust explosion during the milling step of the tableting process. Milled amoxicillin trihydrate with diameters less than 500 μm will mix with air to form a dust-air mixture capable of producing a dust explosion. There are multiple ignition sources for an explosion to occur, including the heat generated due to friction during milling, dispersion of static electricity created during operations, and electrical components. Dust explosions can occur suddenly and cause several secondary explosions, making mitigation necessary for employee safety.

Due to the batch nature of this process, an employee will likely be present in the area and other employees will be on site. This creates a high likelihood that if this event would occur, an employee fatality would be likely, which would categorize the severity as catastrophic. The likelihood of a dust explosion occurring is improbable but not impossible. There have been no

reported cases of an amoxicillin trihydrate or similar API dust explosions, but there is a chance that it will occur. Utilizing the Crawl and Louvar risk matrix in figure 6.1.4-1, the severity and the likelihood place the risk at risk level C, which is an acceptable risk as long as safeguards are put in place to reduce risk to as low as reasonably practical. Due to the catastrophic severity, the target mitigated event frequency is 1×10^{-6} (Crawl et al., 2019). Due to the improbable nature of this event, the unmitigated event frequency is assumed to be once every 1000 years, or 1×10^{-3} .

Risk Matrix					Likelihood			
1. Select the severity from the highest box in either of columns 1, 2, or 3. Read the Category and Safety Severity Level from the same row. 2. Select the likelihood from columns 4 through 7. 3. Read the Risk Level from the intersection of the severity row and the likelihood column. TMEF: Target mitigated event frequency (yr ⁻¹). TQ: Threshold quantity—see Table 1-15.					4 LIKELY	5 UNLIKELY	6 IMPROBABLE	7 IMPROBABLE, BUT NOT IMPOSSIBLE
					Expected to happen several times over the life of the plant	Expected to happen possibly once over the life of the plant	Expected to happen possibly once in the division over the life of the plant	Not expected to happen anywhere in the division over the life of the plant
					0–9 years	10–99 years	≥ 100 years	> 1000 years

Figure 6.1.4-1. Risk Matrix

Multiple methods can be implemented to reduce the likelihood and severity of a dust explosion. Reducing the amount of ignition sources is critical to reducing the likelihood of a dust explosion occurring. Utilizing grounding and bonding techniques can reduce the likelihood of the buildup of static charge due to friction within the process. Additionally, the introduction of independent layers of protection (ILP) reduces the frequency of an event occurring. Without any mitigation, the frequency needs to be reduced by 1×10^{-3} . A blast wall around the control room, a passive ILP, would count for 1×10^{-2} . While the product needs to be sterile, an inherently safer

design would ensure that any section with dust would not be enclosed more than necessary to maintain sterility. Inherently safer design, also a passive ILP, would count for an additional 1×10^{-2} ILP credits. When both of these are implemented, the frequency reduces to 1×10^{-7} .

Reducing the severity can be accomplished by minimizing the amount of dust that can accumulate. Implementing a robust dust collection system would prevent large accumulations of the dust. If an explosion were to occur, minimizing the amount of dust available to explode and therefore reducing the available energy, would result in a less severe explosion compared to a system with an undersized or no dust collection.

6.2 Quality Assurance and Quality Control

6.2.1 Overview of Relevant Regulations

Because we are manufacturing in Australia, we are required to follow Australian law. This primarily involves following requirements of the Therapeutic Goods Administration (TGA) and their Pharmaceutical Inspection Co-operation Scheme (PIC/S) guide to current Good Manufacturing Practice (cGMP). Part I of the guide covers finished medicinal products (here, tablets), while Part II covers API production, and the 20 annexes cover specific situations.

Australian policy is also tightly aligned with guidelines from the ICH, a body designed to harmonize pharmaceutical requirements across countries. In particular, we must take note of ICH Q8 (Pharmaceutical Development), Q9 (Quality Risk Management), Q7 (Good Manufacturing Practice), Q2 (Analytical Validation), and Q1 (Stability Testing). Additionally, all manufacturing will be done in a cleanroom. This cleanroom will be designed according to ISO 14644 standards.

6.2.2 Quality Target Product Profile

This section outlines our quality target product profile (QTPP), or a summary of the quality characteristics of our tablets. We will also include critical quality attributes (CQAs), or properties that must be within a specific range to ensure desired product quality. These are articulated in ICH Q8 on Pharmaceutical Development, and form the basis of Quality by Design (QbD).

Below is the list of QTPP elements and CQAs for our amoxicillin tablets:

- Dosage form: Tablet
- Dosage strength: 500 mg
- Stability: 36 months at room temperature
- Assay, content uniformity
- Tablet dissolution
- Impurity profile

- Residual solvents (methanol, ethanol), water content
- Microbial testing (total aerobic microbial count, total yeast/mold count)

6.2.3 Quality Assurance Systems

Quality Assurance (QA) systems ensure that our product meets our QTPP set above. Specific QA systems that our process will incorporate include a batch release system, change control system, data integrity policies, and standard operating procedures (SOPs).

Most of our processing is continuous, but the final step of our process (drying) is in batch. So, batches of API can still be tested before they are sent forward into the tableting process. Batches will be tested for a number of CQAs before being released, as listed above. Batch release is contingent on meeting all requirements. If there is a deviation, it must be investigated, which will be difficult given the continuous nature of our process. However, there will also be a number of in-process controls of our CPPs. We do not have a procedure for re-working batches, so batches out of specification will be discarded. In the future, we may develop a process for re-working to minimize waste. Batch release of tablets is also contingent on meeting specifications. These specifications will mostly be tablet-specific, but they will also include some CQAs of the API, in case there is contamination in the tableting process itself.

A change control system is essential to making sure all changes are documented, and that the quality team is aware of any possible changes. These changes may include raw materials, operating conditions, equipment changes, or updates to QA and QC systems. Any changes must be proposed, assessed for risk (in accordance with ICH Q9), and approved by QA before being documented and implemented. This is essential for maintaining the safety of the product, and in case of auditing.

Data integrity is critical for ensuring accuracy and safety, as well as in cases of auditing. The facility will follow ALCOA+ principles of data integrity, including user verification, controlled data entry, review and approval, and secure data storage. All data recording will be electronic to ensure there are no errors in transferring data from paper to electronic; therefore, no data should be written on paper, as it will not follow ALCOA+ principles.

All processes will be written in an SOP to ensure that procedures are well-documented, and all employees will be trained and understand all relevant SOPs to their work. Any changes to SOPs must go through the change control process and will be reviewed and approved by QA. SOPs are critical to maintaining cGMP.

6.2.4 Quality Control Systems

Quality control (QC) systems sample and test materials to ensure compliance with our QTPP. They operate under the guidelines discussed above, and test raw materials, conduct in-process controls, and batch verification, as well as handle out-of-specification results.

The PIC/S guide covers testing of raw materials in Annex 8 (p. 163-164). It stipulates that each container of starting materials must be sampled, unless the manufacturer has been validated. We plan to validate each manufacturer of starting materials according to Annex 8, Section 3, by ensuring that there are regular audits of the manufacturer's QA system conducted by us or a local regulatory agency, and ensuring that we are aware of the process that the manufacturer of the starting materials is taking. However, we still must take a representative sample of starting materials and validate that they meet specifications.

QC is also responsible for in-process controls. These are performed during the continuous process and monitor the process to ensure that CQAs are being met. This ensures that issues can

be detected before a batch is out-of-specification. Additionally, QC verifies each batch of both API and tablets based on CQAs in the QTPP and sends the data to QA for batch approval.

6.2.5 Stability Testing

Existing quality testing ensures that the batches are safe when released, but we also must ensure that the tablets are safe for their full shelf life of three years. So, we must conduct stability testing to ensure that our parameters remain in-specification for the duration of our shelf life.

One batch per month will be selected to be “put on stability”. It will be stored at two conditions: a “standard” condition of 25°C and 40% relative humidity (RH), and an “accelerated” condition of 40°C and 75% RH. It will be tested at the following intervals: 1 month, 3 months, 6 months, 12 months, 18 months, 24 months, and 36 months. So, including the initial testing, 15 samples are needed per batch that is selected.

6.3 Environment

The process produces primarily dilute aqueous waste streams containing residual reactants, byproducts, and trace amounts of antibiotics. Most of the equipment operates at or near atmospheric pressure, with the exception of a single vessel operating at 10 bar; however, no high-pressure emissions or hazardous releases are expected. No particularly harmful gases or solids are generated. While the waste streams are mostly water, proper management is required to prevent environmental impacts on aquatic ecosystems and microbial communities. The primary environmental considerations for this process are therefore the treatment of these wastewater streams and the mitigation of potential antimicrobial resistance from trace antibiotics.

6.3.1 Wastewater Treatment

Waste streams containing excess hydroxyphenylglycine methyl ester (HPGME), 6-aminopenicillanic acid (6-APA), hydrochloric acid (HCl), ammonia, water, ethanol, methanol, and amoxicillin will exit the process and be directed to a wastewater treatment facility. Although these streams are expected to be highly dilute and composed primarily of water, treatment is still required to prevent environmental impacts. Methanol and ethanol are biodegradable organic compounds; however, if discharged untreated they can increase the chemical oxygen demand (COD) of receiving waters. Elevated COD results in microbial degradation that consumes dissolved oxygen, which can harm aquatic organisms (Wang et al., 2017). Ammonia is a major nitrogen pollutant in wastewater and can contribute to eutrophication in natural water bodies, leading to algal blooms and reduced oxygen levels that are harmful to aquatic life (Journal of Environmental Management, 2021). Hydrochloric acid is acidic and can significantly alter pH levels, potentially damaging ecosystems and interfering with biological treatment processes.

Although the amount of HCl used in this process is relatively small, neutralization is still required to ensure safe discharge conditions. The organic compounds HPGME, 6-APA, and amoxicillin may also contribute to organic loading and could potentially impact microbial communities in aquatic ecosystems (Zhang et al., 2025).

To mitigate these environmental concerns, all wastewater streams generated by the process will be sent to a municipal wastewater treatment facility. Treatment will likely involve conventional processes such as activated sludge biological treatment, nitrification and denitrification for nitrogen removal, neutralization for pH control, and other biological treatment methods to degrade organic contaminants before discharge (Zhang et al., 2018).

Typical costs for treating industrial wastewater streams of this type in Australia range from approximately \$1.92 to \$2.84 per kL. Assuming an average treatment cost of \$2.25 per kL, the estimated wastewater treatment expense for this process is approximately \$590.70 per day.

6.3.2 Antibiotic Resistance

Antimicrobial resistance (AMR) occurs when bacteria and other microorganisms evolve mechanisms that render antibiotics ineffective, undermining the treatment of common infections (WHO, 2014). Amoxicillin is an antibiotic compound, and the release of antibiotics into the environment raises concerns regarding the development of AMR (Kümmerer, 2009). Even low concentrations of antibiotics in wastewater can create selective pressure that favors the survival and proliferation of antibiotic-resistant bacteria. These resistant microorganisms can spread resistance genes through microbial communities, potentially impacting human and environmental health (Larsson, 2014). Because trace amounts of amoxicillin may be present in the process wastewater, untreated discharge could contribute to this environmental risk. To

mitigate this concern, all wastewater streams generated by the process will be directed to a wastewater treatment facility. Proper wastewater management therefore plays an important role in minimizing the potential contribution of pharmaceutical manufacturing to antimicrobial resistance in the environment.

6.4 Societal Impact

The societal and ethical implications of establishing this pharmaceutical facility include both community integration and economic development. The plant will provide local employment opportunities, offering fair wages, safe working conditions, and training on production processes and product handling. By fostering workforce development, the facility will contribute to economic growth and skill-building within the community.

In addition, this will be the first dedicated antibiotics production facility in Australia, strengthening the national supply chain for essential medications. The plant is designed to produce sufficient quantities of amoxicillin to meet the needs of the entire Australian population, enhancing public health security and reducing dependence on imported antibiotics.

7. Conclusion

7.1 Viability of the Process

We believe that the process described in this report is an effective process for producing amoxicillin at scale. Additionally, we believe that the process as described is reasonably optimized, especially given our yield of reacted amoxicillin is around 82%.

This process is also economically viable. We report an IRR of 26%, which indicates a strong possibility of profitability. However, most of our economic analysis is done using only order-of-magnitude estimates, such as the Lang factor, correlations for equipment and changing materials, and rough estimates of labor costs. Additionally, the cost of raw materials fluctuates, especially given the supply chain uncertainty inherent to production on the Australian continent.

So, instead of fully recommending the project, we recommend further economic analysis of the process, particularly in the areas of pharma-specific items such as cleanroom, air filtering, and QA/QC costs. Once this is done, we will be able to more accurately estimate the capital and operating costs, and give a more thorough assessment of the viability of the project, though we are confident its viability is strong.

7.2 Recommendations for Future Work

However, we also have a number of recommendations at the future design stage for additional process optimization or economic benefit. First, we recommend an economic analysis of in-house production of 6-APA. This raw material alone makes up 25% of our cash cost of production. Additionally, its price is not stable, and while we conducted a sensitivity analysis on the price of 6-APA, the goal of this project is to increase the supply chain independence of

Australia, and while the current process is a good first step, internal production of 6-APA would further the goal of supply chain independence, especially in cases of large-scale disruptions.

Our second recommendation is to look further into recycle streams. There are two large opportunities to recycle in this process: first, water permeating through the reverse osmosis membrane can be sent back into the first reactor, and second, ethanol leaving the rotary drum filter can be recycled for the next washing step. These processes were not assessed at this time, due to the low expense of ethanol (\$15,000 USD per year) and pharma-grade water (\$26,000 USD per year), but this is an easy way to cut costs and use less material, also substantially reducing waste costs.

Third, we should more accurately estimate costs inherent to pharmaceutical production, such as cGMP, QA/QC, and compliance costs. These costs are accounted for in labor, but additional capital costs are required, such as cleanrooms and air filters, that our order-of-magnitude estimates do not account for. Future design work should design these processes in-depth and ensure they are accurately costed.

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Professor Ron Unnerstall

Research Librarian Shannon McCullough

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10. Appendices

10.1 Kinetic Parameters and Rate Laws used for Reactor Design

Table 10.1-1. Kinetic parameters used for reactor design.

Parameter	Meaning	Value	Unit
k_{cat1}	Catalytic constant for HPGE consumption	0.153	$\mu\text{mol IU}^{-1} \text{min}^{-1}$
k_{cat2}	Catalytic constant for amoxicillin hydrolysis	0.198	$\mu\text{mol IU}^{-1} \text{min}^{-1}$
K_{m1}	Michaelis constant for HPGE	11.816	mM
K_{m2}	Michaelis constant for amoxicillin hydrolysis	2.683	mM
T_{max}	Maximum enzymatic conversion	0.761	Dimensionless
K_{en}	6-APA adsorption constant to enzyme	6.732	mM
k_{AB}	Inhibition constant for HPGE	3.780	mM
k_{AN}	Inhibition constant for amoxicillin	9.174	mM
k_{AOH}	Inhibition constant for p-hydroxyphenyl glycine	16.153	mM
k_{NH}	Inhibition constant for 6-APA	62.044	mM

Amoxicillin Synthesis: $v_s = \frac{k_{cat1} C_E C_{AB} C_{NH} T_{max}}{(K_{EN} + C_{NH})(K_{m1} (1 + \frac{C_{AN}}{k_{AN}} + \frac{C_{NH}}{k_{AOH}} + \frac{C_{AOH}}{k_{AOH}}) + C_{AB}}$; where C_E is concentration

of enzyme, C_{AB} is concentration of HPGME, C_{NH} is concentration of 6-APA, C_{AN} is concentration of amoxicillin and C_{AOH} is concentration of methanol

Amoxicillin Hydrolysis: $v_{H_2} = \frac{k_{cat2} C_E C_{AN}}{(K_{m2} (1 + \frac{C_{AB}}{k_{AB}} + \frac{C_{NH}}{k_{NH}} + \frac{C_{AOH}}{k_{AOH}}) + C_{AN}}$; where C_E is concentration of

enzyme, C_{AN} is concentration of amoxicillin, C_{AB} is concentration of HPGME, C_{NH} is concentration of 6-APA, and C_{AOH} is concentration of methanol.

10.2 Calculations for Solubilities in Ethanol Solutions

12% EtOH Washing liquid at 20°C:

Amoxicillin Solubility:

$$5.39 \frac{\text{mol Amox}}{\text{kg solvent}} \times \frac{419.45 \text{ g Amox}}{1 \text{ mol Amox}} \times \frac{1 \text{ kg solvent}}{1000 \text{ g solvent}} \times \frac{1}{10^3} = 0.00226 \frac{\text{g Amox}}{\text{g solvent}}$$

HPGME Solubility:

$$0.1578 \frac{\text{mol HPGME}}{\text{mol solvent}} \times \frac{181.19 \text{ g HPGME}}{1 \text{ mol HPGME}} \times \frac{1 \text{ mol solvent}}{19.44 \text{ g solvent}} \times \frac{1}{10^3} = 0.00147 \frac{\text{g HPGME}}{\text{g solvent}}$$

12% EtOH Washing Liquid at 25°C

Amoxicillin Solubility:

$$6.44 \frac{\text{mol Amox}}{\text{kg solvent}} \times \frac{419.45 \text{ g Amox}}{1 \text{ mol Amox}} \times \frac{1 \text{ kg solvent}}{1000 \text{ g solvent}} \times \frac{1}{10^3} = 0.00270 \frac{\text{g Amox}}{\text{g solvent}}$$

HPGME Solubility:

$$0.22635 \frac{\text{mol HPGME}}{\text{mol solvent}} \times \frac{181.19 \text{ g HPGME}}{1 \text{ mol HPGME}} \times \frac{1 \text{ mol solvent}}{19.44 \text{ g solvent}} \times \frac{1}{10^3} = 0.00213 \frac{\text{g HPGME}}{\text{g solvent}}$$

30% EtOH Washing Liquid at 20°C

Amoxicillin Solubility:

$$6.34 \frac{\text{mol Amox}}{\text{kg solvent}} \times \frac{419.45 \text{ g Amox}}{1 \text{ mol Amox}} \times \frac{1 \text{ kg solvent}}{1000 \text{ g solvent}} \times \frac{1}{10^3} = 0.00266 \frac{\text{g Amox}}{\text{g solvent}}$$

HPGME solubility:

$$0.3865 \frac{\text{mol HPGME}}{\text{mol solvent}} \times \frac{181.19 \text{ g HPGME}}{1 \text{ mol HPGME}} \times \frac{1 \text{ mol solvent}}{22.04 \text{ g solvent}} \times \frac{1}{10^3} = 0.003177 \frac{\text{g HPGME}}{\text{g solvent}}$$

30% EtOH Washing Liquid at 25°C

Amoxicillin Solubility:

$$7.61 \frac{\text{mol Amox}}{\text{kg solvent}} \times \frac{419.45 \text{ g Amox}}{1 \text{ mol Amox}} \times \frac{1 \text{ kg solvent}}{1000 \text{ g solvent}} \times \frac{1}{10^3} = 0.00319 \frac{\text{g Amox}}{\text{g solvent}}$$

HPGME Solubility:

$$\cancel{0.004388} \frac{\text{mol HPGME}}{\text{mol solvent}} \times \frac{181.19 \text{ g HPGME}}{1 \text{ mol HPGME}} \times \frac{1 \text{ mol solvent}}{22.04 \text{ g solvent}} \times \frac{1}{10^3} = 0.004388$$

* HPGME solubilities were approximated by using the midpoint for linear interpolation between the nearest ethanol concentrations.

10.3 System of Linear Equations to Solve for Required Amoxicillin Flows Into Washing

$m_{\text{Amox}} = 105.67m_{\text{HPGME}} \rightarrow$ Determined by inlet flow ratio of Amoxicillin to HPGME

$m_{\text{Wash}} = 227.89m_{\text{HPGME}} \rightarrow$ 1/HPGME solubility w/w to remove all HPGME

$m_{\text{Amox}} - 0.00319m_{\text{Wash}} = 3.17 \rightarrow$ Amoxicillin in - Amoxicillin losses due to solubility = 3.17 g/s

out for 11.4 kg/h product to drying

10.4 Raw Material Pricing

Below in Table 10.4-1, we include the material, rate of consumption, cost per unit, and cost per year, along with sources, as discussed in Section 5.2.1.

Table 10.4-1. Raw material cost calculation.

Material	kg/h	kg/yr	\$/kg	\$/yr	Source
6-APA	7.6	54,720	\$36.56	\$2.00 M	[1]
HPGME	6.5	46,800	\$11.07	\$518 K	Table 10.4-2
HCl	1.32	9,504	\$0.18	\$1.72 K	Table 10.4-3
EtOH	2.25	16,227	\$0.95	\$15.4 K	[2]
NH ₃	0.62	4,464	\$0.45	\$2.01 K	[3]
EG	N/A	22,710	\$0.60	\$13.6 K	[4]
Cellulose	1.07	7,695	\$7.07	\$54.4 K	[3]
Mg stearate	0.13	945	\$16.93	\$16.0 K	[2]
Povidone	0.75	5,400	\$15.44	\$83.4 K	[2]
Bottles and lids		8,000,000	\$0.18	\$1.44 M	[5]

[1] indiapharmaoutlook.com/news

[2] zauba.com

[3] worldbank.org

[4] procurementresource.com

[5] fhpkg.com

While unit prices could not be found and sourced for HPGME and HCl, import data was found from eximtradedata.com (for HPGME) and volza.com (for HCl). To compute a unit price, we calculated the unit cost (per kg) of the 10 most recent shipments as of March 2026, neglecting outliers. These samples are included in Table 10.4-2 (for HPGME) and Table 10.4-3 for (HCl).

Table 10.4-2. Shipment sample of HPGME to calculate unit price.

Quantity (kg)	Cost (USD)	Cost per kg
16000	\$186,149.58	\$11.63
10000	\$113,814.28	\$11.38
16000	\$186,149.58	\$11.63
242	\$2,404.37	\$9.94
16000	\$158,966.64	\$9.94
15658	\$156,562.27	\$10.00
16000	\$185,595.57	\$11.60
40000	\$453,902.20	\$11.35
1400	\$16,239.61	\$11.60
30600	\$354,951.52	\$11.60

Table 10.4-3. Shipment sample of HCl to calculate unit price.

Quantity (kg)	Cost (USD)	Cost per kg
20700	\$3,098.79	\$0.150
20700	\$5,334.92	\$0.258
19200	\$3,994.99	\$0.208
20700	\$5,363.51	\$0.259
41400	\$5,990.99	\$0.145
19200	\$4,146.03	\$0.216
62100	\$8,986.49	\$0.145
20700	\$2,995.50	\$0.145
41400	\$5,990.99	\$0.145
62100	\$8,986.49	\$0.145